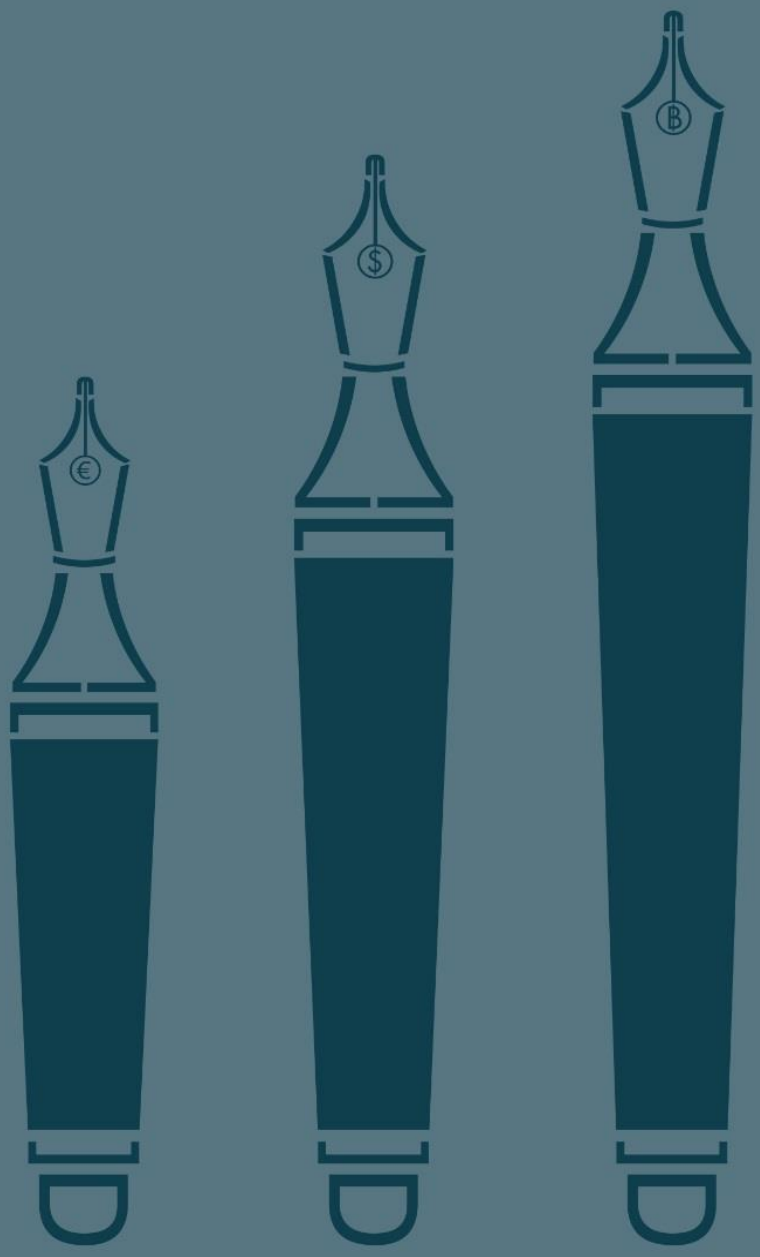
BANK OF THAILAND

MONETARY POLICY REPORT DECEMBER 2016





BANK OF THAILAND

Monetary Policy Report

December 2016

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
December 2016

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Jamlong Atikul	Member
Mr. Porametee Vimolsiri	Member
Mr. Apichai Boontherawara	Member
Mr. Sethaput Suthiwart-Narueput	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for 2016 and for the medium term. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

Monetary Policy Report

December 2016

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy is projected to expand by 3.2 percent in 2016 and 2017 driven mainly by private consumption, public spending, and exports of services. The current estimates are close to the projections in the previous *Monetary Policy Report*. Key drivers include: (1) expansion in private consumption supported by improved farm income, (2) continued fiscal stimulus, and (3) recovery in merchandise exports. These positive developments offset (1) the slowdown in exports of services that was a result of recent government measures to curb illegal tour operators and subdued tourism activities during the mourning period, (2) low private investment, and (3) the lower-than-expected momentum from government consumption. Nevertheless, the Thai economy faces increased downside risks and greater uncertainties from both domestic and external factors.

Headline inflation projection for 2016 and 2017 is revised down from 0.3 and 2.0 percent to 0.2 and 1.5 percent, respectively. This downward revision reflects the subdued inflationary pressures from the gradual recovery in demand and lower fresh-food prices despite rising oil prices. The Committee thus expects headline inflation to return to the lower bound of the target band within the first quarter of 2017. Meanwhile, core inflation is revised down given a slower rise in prices of food purchased for consumption at home and away from home.

The Monetary Policy Committee decided to keep the policy rate on hold at both meetings in November and December 2016. According to their assessment, the overall economic and inflation outlook was largely unchanged from the previous quarter as the economic recovery remains on track with headline inflation on the rise. Meanwhile monetary conditions remain accommodative and conducive to the economic recovery. Financial stability remains sound, although there are pockets of risks that continued to warrant close monitoring. Nonetheless, the Thai economy would still be facing a number of uncertainties going forward, and thus the Committee will closely monitor risk developments and their impact on the Thai economy, and conduct monetary policy in an appropriate and timely manner to support the ongoing recovery without contributing to unnecessary accumulation of fragilities in the financial system.

1.1 Growth and inflation prospects

The Committee maintains the GDP growth forecast for 2016 and 2017 at 3.2 percent (Table 1.1). Key growth drivers include private consumption, public spending, and exports of services. The decision to maintain the forecast is due to (1) higher-than-expected private consumption given improved farm income thanks to higher agricultural output and rubber prices, (2) the slight recovery in export receipts, and (3) additional fiscal stimulus. These positive developments help offset (1) the slowdown in exports of services, (2) low private investment, and (3) lower-than-expected growth in public spending especially in government consumption.

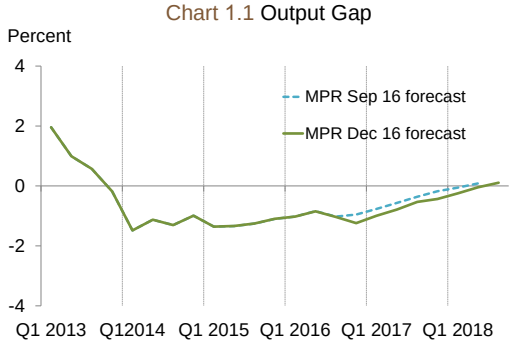
Table 1.1 Forecast summary

Percent	2015*	2016	2017
GDP growth	2.8	3.2	3.2
		(3.2)	(3.2)
Headline inflation	-0.9	0.2	1.5
		(0.3)	(2.0)
Core inflation	1.1	0.7	0.8
		(0.8)	(1.0)

Note: *Outturn
() September 2016 MPR
Source: Office of National Economic and Social Development Board, Ministry of Commerce, calculations by Bank of Thailand

Regarding the inflation outlook, as the economy is expected to expand in 2016 and 2017 at a rate close to the previous assessment, **demand-pull inflationary pressures remain largely unchanged,** slowly increasing in tandem with the closing of the output gap **(Chart 1.1)**. Meanwhile, **cost-push inflationary pressures has softened** given the significant decline in fresh food prices. This is particularly the case for the prices of rice as well as fruits and vegetables which declined after the drought subsided and expected to return to normal levels earlier than expected. Meanwhile, the unexpected pickup in oil prices has pushed up production costs of goods and services domestically, albeit not enough to offset the fall in fresh food prices. In addition, lower prices of fresh food is likely to slow down the increase in prices of food purchased for consumption at and away from home in the period ahead. **The Committee therefore revises the core inflation forecast for 2016 and 2017 down from 0.8 and 1.0 percent, respectively, to 0.7 and 0.8 percent. The headline inflation forecast for 2016 and 2017 is also revised down from 0.3 and 2.0 percent to 0.2 and 1.6 percent, respectively.** Nonetheless, the Committee expects headline inflation to return to the lower bound of the target band within the first quarter of 2017, a slight delay from the fourth quarter of 2016 as reported in the previous *Monetary Policy Report*.

Chart 1.1 Output Gap



The Committee has incorporated key economic developments into the growth and inflation forecasts as summarized below.

(1) Trading partners' economies are likely to expand at a pace close to the previous assessment (Table 1.2), but face greater downside risks and uncertainties.

Table 1.2 Growth assumptions for Thailand's trading partners

Percent (%YoY)	Weight (%)	2015	2016		2017	
			Jun 16	Sep 16	Jun 16	Sep 16
United States	14.9	2.6	1.5	1.6	2.2	2.2
Euro area	10.0	1.6	1.6	1.6	1.2	1.1
United Kingdom	2.5	2.3	1.6	2.0	0.6	0.8
Japan	13.6	0.6	0.5	1.0	0.9	1.1
China	15.7	6.9	6.5	6.6	6.3	6.3
Asia (ex Japan and China)*	37.4	3.5	3.2	3.3	3.5	3.5
Total**	100	3.3	2.9	3.0	3.1	3.1

Note: * Weighted by each trading partner's share of Thailand's total exports in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8%), Taiwan (2.5%), Indonesia (5.9%), Korea (2.8%), and the Philippines (3.7%)
** Weighted by each trading partner's share of Thailand's total exports as of 2014 (13 countries)

The Committee has made a slight upward adjustment to trading partners' growth projection for 2016 to account for recent data releases for the third quarter of 2016 that turned out better than expected. Several economies are projected to record higher growth compared with the previous assessment due to (1) a softer-than-expected impact of Brexit on the United Kingdom (UK) economy, (2) a better export growth in Japan, and (3) a gradual improvements in exports of Asian economies (excluding Japan and China) that would bolster private consumption.

For 2017, the projection for trading partners' GDP growth is maintained at 3.1 percent. The Committee projects trading partners to gradually recover notwithstanding a higher growth base in 2016. **Nonetheless, growth momentum is likely to soften somewhat in tandem with Asian economies which are expected to recover at a slower pace** given increased uncertainties pertaining to the U.S. trade policies that might incline toward greater protectionism under the new administration. This also includes a potential collapse of the Trans-Pacific Partnership (TPP) which could defer business investment among member countries. However, major advanced economies are expected to recover gradually supported by improving private consumption.

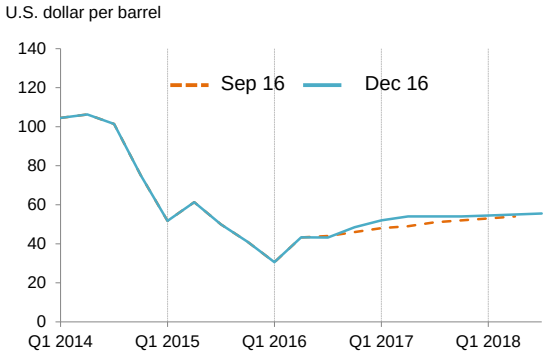
Monetary policies in major advanced economies remain accommodative with an exception of the U.S. The Bank of England (BOE) has continued to purchase government and corporate bonds alongside the use of Term Funding Scheme (TFS). Meanwhile, the European Central Bank (ECB) has extended its quantitative easing (QE) program from until March to until December 2017. The Bank of Japan (BOJ) maintains its negative interest rate policy and quantitative and qualitative easing (QQE) with yield curve control. On the contrary, the Federal Reserve raised the federal funds rate up by 0.25 percent on December 14, 2016 and revised its dot plot to reflect a faster normalization path from two to three hikes in 2017.

The outlook for trading partners' economies possesses a larger degree of uncertainties, especially from the impact of the U.S. economic policies under the new administration that remain unclear in many aspects including trade , immigration, and tax cuts. Such uncertainties along with monetary policy divergence in major advanced economies would add to volatility in global financial markets and international capital flows by more than previously expected. Moreover, to account for improved investors' confidence on U.S. economic growth and the Federal Reserve's normalization path into the exchange rate assumption, the Committee expects Asian currencies (excluding the renminbi) to be weaker than the previous assessment.

Risks to growth in trading partners' economies are tilted toward the downside. Key risks and uncertainties are (1) uncertainty in monetary policies of major advanced economies, (2) uncertainty in the U.S. economic policies, (3) uncertainty pertaining to the post-Brexit trade and investment negotiation between the UK and EU, (4) political uncertainties in Europe that may heighten due to the upcoming elections in several member countries especially France and Germany, and (5) risks in the European and Chinese financial systems. These risk factors will affect volatility in the global financial markets and may have a greater-than-expected impact on the real economy.

(2) Global oil prices have risen faster than previously expected (Chart 1.2), after OPEC has reached an agreement to cut production, resulting in a likely increase in prices of non-oil commodities. The Committee thus revises up assumptions on Dubai prices in 2016 and 2017 from 41.0 and 50.0 U.S. dollars per barrel to 41.4 and 53.5 U.S. dollars per barrel, respectively. The increase is mainly due to the tightened supply after an agreement to cut production was reached among OPEC members in their meeting on November 30, 2016 and among Non-OPEC producers in their meeting on December 10, 2017. **The production cut will prompt global crude prices to reach equilibrium faster. However, higher crude prices might induce higher production of shale oil which would maintain downward pressure on oil prices going forward.** In this regard, crude prices are expected to stabilize in the second half of 2017 when shale oil production resumes and gradually ramp up in response to higher demand given the ongoing global economic recovery.

Chart 1.2 Assumptions on Dubai oil price



In addition, higher oil prices relative to the previous assumption will affect commodity prices. Metal prices face upward pressures as the market expects a recovery in global demand from the U.S. infrastructure investment policies under the new administration.

Going forward, risks that could make global crude oil prices deviate from the baseline projection are balanced. Key downside risks include an earlier-than-expected resumption in shale oil production and lower global demand for oil in the case of a sluggish global recovery. Upside risks come from conflicts in the Middle East that could spread to major production sites. **Furthermore, the Committee views that the outlook of global crude prices remains largely uncertain as a result of supply side factors**, namely (1) market concerns over the implementation of production cut and (2) uncertainty in the U.S. economic and energy policies that can add further volatility.

(3) **The number of tourists has declined more than previously assessed**

due to measures to curb illegal tour operators and limited festivities during the mourning period for the late King.

The government's measure to curb illegal tour operators has had a stronger impact on tourism than the previous assessment, according to which the impact would be limited in the short term to only some businesses. In the period ahead, the increase in the prices of package tours to Thailand may prompt price-sensitive Chinese tourists to travel to other destinations. Moreover, tourism during the mourning period has muted slightly due to the absence of public celebrations during the first 30 days, although the situation is expected to improve in early-2017.

At the same time, the government has issued new measures to support the tourism sector, including (1) a temporary discount on visa-on-arrival fees by 1,000 baht for tourists from 19 countries between December 2016 and February 2017, (2) an extension to the permitted periods of stay in Thailand for tourists on long-stay visas, and (3) an effort to foster mutual understanding and confidence among Chinese tour operators and tourists. The Committee expects the first two measures to bolster the number of foreign individual tourists. **The Committee thus revises the projected number of foreign tourists for 2016 and 2017 down from 33.6 and 36.3 million to 32.4 and 34.1 million, respectively.**

(4) Farm income has improved on the back of both prices and output. Prices of agricultural goods have increased faster than previously assessed, especially for rubber after output declined due to floods in the Southern region. Meanwhile, output of other crops has improved after the drought subsided. Going forward, agricultural output is expected to receive a boost from the new water management plan for dry season in 2017, which includes higher water allocation for farming compared with the previous year as well as measures to encourage production of corn instead of second-crop rice in 2 million *rais*. The increased output will put downward pressure on prices of some products such as rice. However, the Committee sees that farm income would not decline significantly as the government has put forward additional measures to subsidize income for farmers including cash transfers to rubber farmers (11.5 billion baht), rice farmers (31.5 billion baht), and low-income farmers (6.5 billion bath), as well as measures to support farmers who grow white rice, Pathumthani fragrant rice, and glutinous rice in addition to jasmine rice (9 billion baht).

Given the developments above, the forecasts for growth and inflation can be summarized as follow:

(1) While the volume of merchandise exports has improved from the previous assessment, it is expected to recover slowly. Improvements in export growth are limited to some industrial products that have benefited from the relocation of production base and expansion into new markets, namely (1) electrical appliances due to increased foreign demand especially from Europe and the U.S., (2) automobile and parts due to (a) relocation of pneumatic tires production bases from China to Thailand and

(b) an expansion into new markets by car manufacturers, and (3) electronics due to (a) an increase in demand from the U.S. for integrated circuits used in the production of smart phones and (b) hard disk drives that gained from the consolidation of production bases from other countries into Thailand to reduce costs (details in Chapter 2). **However, merchandise exports are still expected to grow slowly given the gradual recovery of trading partners' economies, and structural issues pertaining to global trade as well as Thailand's manufacturing production that would take time to resolve.**

Given the slight improvement in the export volume and higher-than-expected prices of oil-related exports due to rising crude oil prices, **the Committee adjusts the projections for export growth for 2016 and 2017 from a contraction of 2.5 and 0.5 percent to an expansion of 0.6 and 0.0 percent, respectively.**

Exports of services are expected to grow at a slower pace than in the previous assessment, especially in the fourth quarter of 2016 and the first half of 2017 due to the lower-than-expected number of foreign tourists. Nonetheless, additional measures from the government to support tourism and to strengthen confidence and facilitate adjustments on part of the tour operators that cater to Chinese tourists will help exports of services recover in 2017.

Imports of goods and services are higher compared with the previous assessment. Imports of goods have increased in line with improved export growth and rising prices of oil and oil-related products. The projection of import of goods for 2016 and 2017 is therefore revised from a contraction of 6.6 and an expansion of 5.6 percent to a contraction of 5.0 percent and an expansion of 7.8 percent, respectively.

Overall, the **current account** in 2016 has registered a slightly larger surplus of 42.2 billion U.S. dollars compared with the previous assessment at 40.4 billion. Meanwhile, the current account surplus in 2017 is expected to record 26.9 billion U.S. dollars, smaller than the previous forecast of 31.8 billion. The revision reflects a decline in receipts from exports of services and an increase in the value of imports of goods.

(2) Private consumption has recorded a larger improvement than previously assessed, partly due to actual data released for the third quarter of 2016 that turned out better than expected, and is expected to continue expanding. Key supporting factors are increasing farm income, rising export prices, and, in part, from the lower debt repayment burden in 2017, as car loans under the first-car tax-rebate scheme are due after five years.

Furthermore, **private consumption has received additional support from government measures** such as tax deduction incentives at the end of 2016 to stimulate consumption and cash transfers for low-income earners. The Committee expects these measures to boost private consumption in the short term and support purchasing power of low-income households during periods when the economy has not fully recovered.

(3) Private investment remains subdued and will recover at a slower pace due to a low level of export of goods, the slowdown in exports of services, and greater uncertainties faced by both the global economy and the Thai economy. Nonetheless, investment by some businesses, such as retail and telecommunication, is expected to continue in tandem with demand expansion. In 2017, government policies will provide a great support to private investment through public-private partnership (PPP) and additional budget for village funds and urban communities. [1] The latter is designed to provide low-cost funding for investment that will support private investment in provincial areas.

(4) Budget disbursement for public spending has been lower than previously expected especially for public consumption, although public investment is likely to be higher than the previous estimate. Actual data for public consumption in the third quarter of 2016 indicate a decline due to lower disbursement efficiency and lower-than-expected disbursement for social-welfare transfers. Going forward, constraints on budget disbursement are expected to remain, while the government's plan to allow private companies to manage health benefits for public-sector employees would improve spending efficiency, but could restrain growth in government consumption.

At the same time, **public investment** in 2016 expands at a slightly slower pace from a delay in SOE budget disbursement, but is expected to pick up in 2017 given the Cabinet's recent approval for new projects and stimulus measures. One of the measures is the 100 billion baht Pracharat projects that encourage investment for social and economic developments in provincial areas. The program aims to provide additional support for existing public investment. Another measure is a program whereby the government matches funding for investment projects between the central government and local governing agencies.[2]

(5) **Demand-pull inflationary pressures** remain largely unchanged in line with the growth forecast that was unchanged from the previous assessment. **Cost-push inflationary pressure** declined from the previous assessment because, despite higher oil prices, fresh food prices have rapidly declined since September due to higher agricultural output after the drought subsided. In the period ahead, prices of fresh food will likely decelerate and return to the normal level. The lower costs of fresh food also mean lower prices of food for consumption both in and away from home. In this regard, **forecasts for both core inflation and headline inflation are revised down.** The forecasts for core inflation in 2016 and 2017 are 0.7 percent and 0.8 percent, respectively, down from 0.8 and 1.0 percent

[1] The program is a part of the mid-year additional budget for 2017 according to the Cabinet's decision on December 7, 2016. Total spending according to this program will be 190 billion baht: (1) 100 billion baht for the strengthening and sustainable economy projects at the provincial level (Pracharat projects), (2) 62,922 million baht for village funds and urban communities, and (3) 27,078 million baht for compensation of fiscal expenses.

[2] According to the Cabinet's decision on September 13, 2016, the program aims to incentivize local governing agencies to utilize savings to develop economic and social infrastructure in the local area.

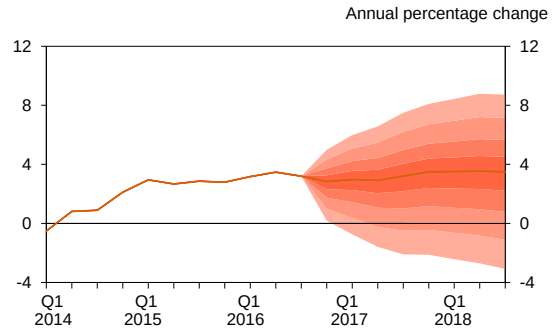
in the previous assessment. Headline inflation for 2017 is projected at 0.2 percent and 1.7 percent, down from 0.3 percent and 2.0 percent, respectively.

Risks to Growth and Inflation Forecasts

The Committee views that the balance of risks to growth continue to be tilted to the downside as reflected in the growth forecast fan chart that is skewed downward throughout the forecast horizon more than the previous *Monetary Policy Report* **(Chart 1.3).** One important **downside risk** to growth is the uncertainty surrounding U.S. trade policies that will have significant implications on confidence for trade and investment. This, in turn, will affect the economic outlook for Thailand's trading partners. Other **ongoing downside risks** include (1) risks in the Chinese financial sector that may lead to slower growth for the Chinese economy and (2) the number of foreign tourists that may be lower than the baseline projection due to measures to curb illegal tour operators. On the other hand, there are some **upside risks** to growth from a faster-than-expected disbursement of government stimulus programs or the implementation of additional stimulus measures. In addition, the U.S. economy may expand at a faster rate than in the baseline scenario due to a stimulus package launched by the new president.

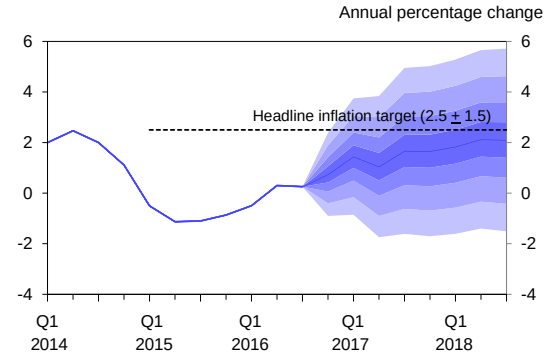
With regard to inflation, **the Committee judges the balance of risks on both headline and core inflation forecasts to be tilted to the downside** in line with the balance of risks to growth. In addition, the variance of forecast estimates is higher than the previous assessment due to uncertainties in oil prices and the economic outlook **(Charts 1.4 and 1.5).**

Chart 1.3 GDP growth forecast



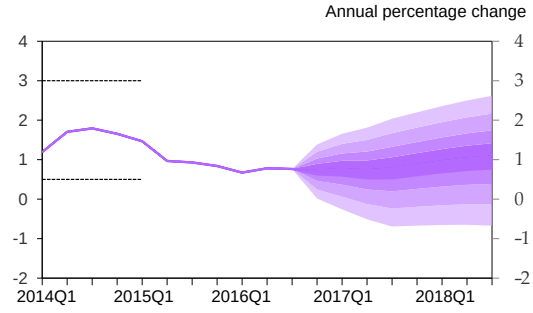
Note: Fan chart covers 90 percent of probability distribution

Chart 1.4 Headline inflation forecast



Note: Fan chart covers 90 percent of probability distribution

Chart 1.5 Core inflation forecast



Note : Fan chart covers 90 percent of proability distribution

1.2 Monetary policy decision

In the fourth quarter of 2016, monetary policy remained accommodative to support the recovery. The outlook of the Thai economy has not changed significantly from the time of meetings in the previous quarter, despite increasing downside risks from both domestic and external factors. The Committee sees that the Thai economy would be facing uncertainties domestically and globally and will closely monitor developments and assess the impact of risks to the economic recovery. The Committee would stand ready to utilize an appropriate mix of monetary policy tools to support Thailand's economic recovery in a timely manner without contributing to financial fragilities to the financial system.

At the **MPC meeting on November 9, 2016**, the Committee voted unanimously to maintain the policy rate at 1.50 percent. The Committee assessed that **the Thai economy continued to expand despite increased downside risks** especially from heightened uncertainties in the global economy. These include political developments abroad and risks to financial stability in Europe and China that could hinder growth in trading partners' economies. The government's measure to curb illegal tours might also cause the number of Chinese tourists to turn out lower than the previous projection. The Committee would continue to monitor the effect of such measure on growth momentum in the tourism sector. **As the Thai economy still faced high uncertainties, the Committee, therefore, affirmed the need to preserve policy space to cushion potential impact should these risks materialize by maintaining the policy rate at the current level. Going forward, key risks are a fragile global recovery and uncertainties in economic and monetary policies of major advanced economies that could add volatility to capital flows and exchange rates.** In order to formulate the appropriate monetary policy, the Committee would then continue to closely monitor risk developments and assess their impact on the Thai economy.

Headline inflation rose slowly, with a higher possibility of returning to the target band later than assessed in the previous *Monetary Policy Report*. The potential delay is caused by a slowdown in recovery of fresh-food prices due to supply factors. **Nonetheless, headline inflation was still expected to gradually pick up, while the public's expectations on the medium-term inflation remained close to the inflation target.** In the period ahead, the Committee assessed that structural changes would keep global inflation relatively low. Such changes include a shift in global oil production that would prevent crude oil prices from accelerating to a high level as in the past. Technological advancement and telecommunication would also reduce prices of goods and services. This inflation outlook is a challenge to central banks around the world (see Box: *Flexible Inflation Targeting in an Evolving Global Economic Landscape*). The Committee would keep a close watch on factors affecting inflation and stand ready to utilize an appropriate mix of availability policy tools in order to ensure that inflation returns to the middle point of the target band and

anchor the public's medium-term inflation expectations.

Monetary conditions remained accommodative and conducive to the economic recovery. Liquidity in the financial system remained high with low borrowing costs as reflected in negative real interest rates. Total corporate financing slowly increased in line with the gradual pace of economic recovery but still concentrated in some industries. **However, the Committee viewed that the recent appreciation of the baht relative to key trading partner currencies might not be beneficial to the economic recovery. While regional currencies including the baht weakened against the U.S. dollar, but the baht depreciated to less compared with overall trading partner currencies** given Thailand's sound economic fundamentals and external stability.

In addition, **the Committee viewed that the conduct of monetary policy under the prolonged low interest rate environment must take into account financial stability considerations.** While financial stability remained sound overall, there are pockets of risks that warranted close monitoring. These included the deterioration of businesses loan quality, particularly among small-and-medium-sized enterprises (SMEs) and households. In addition, the search-for-yield behavior, especially in unrated bonds, might lead to underpricing of risks and therefore warranted close monitoring. The Committee saw these risks as closely connected with the financial system. Thus, the use of policy tools to prevent an accumulation of imbalances in the financial system must be systematically conducted with collaboration across various agencies to limit systemic risks.

At the following meeting on December 21, 2016, the Committee voted unanimously to maintain the policy rate at 1.50 percent. In deliberating this decision, the Committee assessed that **the Thai economy overall continued to expand at a pace close to the previous assessment as key economic drivers remained largely unchanged from the previous meeting.** Although tourism slowed and private investment remained sluggish, negative impacts were offset by improvements in merchandise exports and private consumption. Meanwhile, public spending continued as a major growth engine. **However, downside risks to growth increased** from the possibility of weaker-than-expected trading partners' economies. Trade policies of the new U.S. administration might affect international trade and confidence of the private sector. Moreover, the number of Chinese tourists might turn out lower than previously estimated. In addition, ongoing risks from political developments in Europe and banking concerns in Europe and China still required continued monitoring. **Headline inflation was expected to pick up and return to the target band within the first quarter of 2017** although the timing would largely depend on developments of oil and fresh food prices. Monetary conditions remained accommodative and conducive to the economic recovery, although long-term bond yields rose to the similar level as the average in the previous year following an increase in the U.S. Treasury yields.

In deliberating this decision, the Committee gave due consideration to Thailand's ongoing economic recovery. Although the recent expansion appeared to concentrate in some sectors with higher downside risks, supporting factors for growth momentum continued, especially those from government measures. **Going forward, the Committee assessed that the Thai economy would face heightened uncertainties** from a fragile global recovery and uncertainties in economic and monetary policies in major advanced economies. Particularly, the U.S. policies under the new president that still remained unclear would have significant implications on the pace of recovery of Thailand's trading partners economies and contribute to volatilities in international capital flows and exchange rates. The Committee therefore would continue to closely monitor risk developments and assess their impacts on the Thai economy to formulate appropriate monetary policies to support the recovery of the Thai economy in a timely manner, without adding unnecessary accumulation of financial fragilities to the system.

Going forward, the Committee saw the need for continued accommodative monetary policies and would stand ready to utilize an appropriate mix of available policy tools in order to support the economic recovery and ensure financial stability.

1.3 Appendix: Summary of assumptions and projections

Table 1.3 Forecast for GDP and assumptions			
Percent	2015*	2016	2017
GDP growth	2.8	3.2	3.2
Domestic demand	2.8	2.4	3.2
Private consumption	2.1	3.1	2.6
Private investment	-2.0	-0.6	1.6
Government consumption	2.2	1.1	3.2
Public investment	29.8	9.3	11.9
Exports of goods and services	0.2	1.9	0.6
Imports of goods and services	-0.4	-2.3	2.4
Current account (billion, U.S. dollars)	32.1	42.2	26.9
Value of merchandise exports	-5.6	-0.6	0.0
Value of merchandise imports	-10.6	-5.0	7.8

Note: *Outturns

Table 1.4 Forecast assumptions			
Annual percentage change	2015*	2016	2017
Dubai oil price (U.S. dollar per barrel)	50.8	41.4	53.5
Non-fuel commodity prices (%YoY)	-17.5	-2.6	1.2
Fresh food prices (%YoY)	-10.9	3.9	1.9
Minimum wage in the Bangkok Metropolitan Region (baht per day)	300	300	310
Government consumption (current price) (%YoY) 1/	4.4	3.1	5.9
Public investment (current price) (%YoY) 1/	25.7	8.5	16.1
Fed Funds rate (% at year-end)	0.38	0.63	1.38
Trading partners' economic growth (%YoY) 2/	3.3	3.0	3.1
Regional currencies vis-à-vis the U.S. dollar (Index) 3/	150.7	154.5	160.3

Note: 1/ Including spending on water management plans and infrastructure investment projects
2/ Weighted by each trading partner's share in Thailand's total exports
3/ Appreciation against the US dollar indicated by the minus sign
* Outturns

Table 1.5 GDP growth forecasts by research houses		
	2016	2017
Maybank Kim Eng	3.5	4.5
TISCO Securities	3.4	3.6
Standard Chartered Bank	3.3	3.5
TMB Bank	3.3	3.5
NESDB[2/]	3.3	3.4
Kasikorn Research	3.3	3.3
KT ZMICO Securities	3.3	3.3
NESDB[2/]	3.2	3.0-4.0
Bank of Ayudhya	3.2	3.3
Siam Commercial Bank	3.2	3.3
Phatra Securities	3.2	3.2
BOT	3.2	3.2
Moody	3.2	2.8
UBS	3.1	2.5
Nomura Co Ltd	2.8	2.8
Thanachart Securities	1.5	1.5

Note: Compiled and published by Reuters on December 19, 2016, except:
[1/] Published on October 28, 2016
[2/] Published on November 21, 2016 with the release of GDP data for 2016 Q3
Presented in descending order of 2016 forecasts

Table 1.6 Headline inflation forecasts by research houses		
	2016	2017
Maybank Kim Eng	2.0	2.5
FPO [1/]	0.4	2.1
TMB Bank	0.4	2.0
TISCO Securities	0.3	2.2
Standard Chartered Bank	0.3	1.8
Kasikorn Research	0.2	1.8
Bank Ayudhya	0.2	1.7
Moody	0.2	1.7
Thanachart Securities	0.2	1.7
KT ZMICO Securities	0.2	1.6
Phatra Securities	0.2	1.6
Siam Commercial Bank	0.2	1.6
NESDB [2/]	0.2	1.0-2.0
BOT	0.2	1.5
UBS	0.2	1.4
Nomura Co Ltd	0.1	1.0

Note: Compiled and published by Reuters on September 14, 2016, except:
[1/] Published on July 28, 2016
[2/] Published on August 15, 2016 with the release of GDP data for 2016 Q2
Presented in descending order of 2016 forecasts

Table 1.7 Probability distribution of GDP growth forecast								
Percent	2016	2017				2018		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
10-12	0	0	0	0	1	1	2	3
8-10	0	0	1	3	4	5	6	6
6-8	1	4	7	10	12	13	13	13
4-6	16	21	21	21	22	21	20	19
2-4	49	37	31	26	25	23	22	21
0-2	29	26	24	21	19	19	18	17
(-2)-0	4	9	12	12	11	11	11	11
< (-2)	0	2	4	6	6	7	8	9

Table 1.8 Probability distribution of core inflation forecast								
Percent	2016	2017				2018		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
3.5-4.0	0	0	0	0	0	0	1	1
3.0-3.5	0	0	0	0	0	1	1	1
2.5-3.0	0	0	0	1	2	3	3	4
2.0-2.5	0	1	2	4	6	7	9	9
1.5-2.0	2	7	9	11	13	14	15	16
1.0-1.5	22	23	21	20	20	20	20	19
0.5-1.0	44	33	27	23	22	21	19	18
0.0-0.5	25	23	22	19	18	16	15	14
(-1)-0.0	5	10	12	12	11	10	9	9
(-2)-(-1)	0	2	5	6	5	5	5	5
< -2	0	0	1	2	2	2	2	2

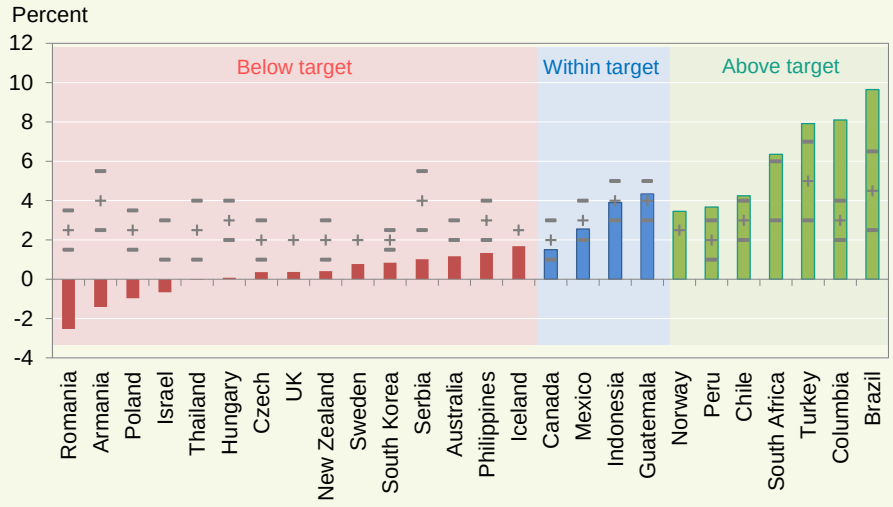
Table 1.9 Probability distribution of headline inflation forecast								
Percent	2016	2017				2018		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3
> 7	0	0	0	0	0	1	1	1
6-7	0	0	0	1	1	1	2	2
5-6	0	0	0	2	2	3	4	5
4-5	0	2	2	6	6	8	9	9
3-4	1	8	7	13	12	14	15	15
2-3	7	21	16	19	18	19	19	18
1-2	29	29	23	21	20	19	18	18
0-1	39	24	23	18	17	16	15	15
(-1)-(0)	20	12	16	12	12	11	9	10
(-2)-(-1)	4	4	8	6	6	6	5	5
(-3)-(-2)	0	1	3	3	3	2	2	2
(-4)-(-3)	0	0	1	1	1	1	1	1
< (-4)	0	0	0	0	0	0	0	0

Flexible Inflation Targeting in an Evolving Global Economic Landscape

The Bank of Thailand has adopted flexible inflation targeting as its main framework for the conduct of monetary policy since 2000. The policy framework is conducive to a disciplined formulation of monetary policy to maintain price stability through clear communication of the monetary policy target, with sufficient flexibility in implementing monetary policy to cope with shocks, thereby allowing the economy to reach potential in a sustainable manner.

However, **monetary policy implementation under the flexible inflation targeting framework has become more challenging**. Recent volatilities in the global economy have caused uncertainties for Thailand's economic recovery and made inflation forecast more difficult. Over the past year, inflation in most inflation-targeting countries were below the targets[1/] **(Chart 1)**. **First, global crude oil prices did not rise as much as in the past** due to structural changes in oil production. Advancements in the shale oil extraction technology allow faster response of oil supply to price movements. At the same time, the global oil demand recovery was subdued consistent with fragile economic conditions. **Second, excess production capacity in the global economy** prevented businesses from increasing prices. **Third, structural shifts in the global economy** such as roles of the rise of technologies that reduce business costs, particularly in service sectors which continue to gain increasing shares in the economy.

Chart 1 Average inflation in 2016 in various countries



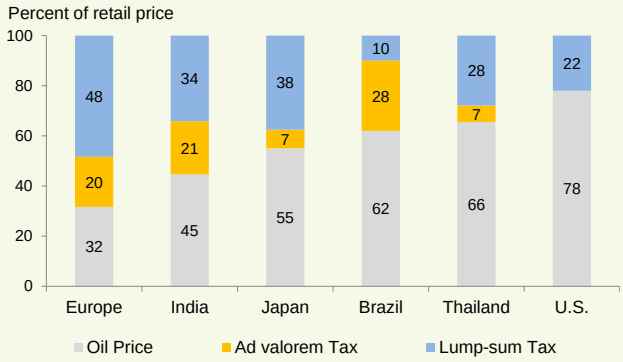
Note: Inflation rates in most countries were below target primarily due to sharp falls in oil prices. Inflation rates that exceeded target were mainly caused by currency depreciation and higher fresh food prices.

Sources: Bloomberg and Calculations by Bank of Thailand (data from January 2016-present)

[1/] Except in some emerging markets where fragile external financial positions caused sharp currency depreciation, leading to higher import prices and above-target inflation.

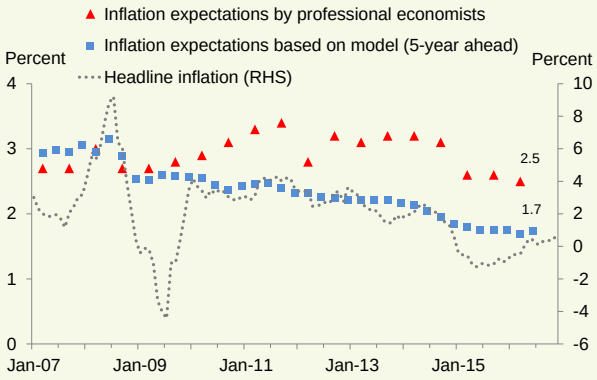
The greater challenge has led inflation targeting central banks to **review the appropriateness of the target and its flexibility under the evolving landscape.** Some central banks such as the Bank of Korea have adjusted their inflation targets downward in line with lower inflationary pressures. However, most central banks have decided to keep their existing targets. In their view, inflation target should be a medium-term objective, and further analyses have to be undertaken to see whether changes in inflation dynamics are due to economic cycles or structural shifts, because changing an inflation target will affect inflation expectations formation.

Chart 2 Retail oil price structures in various countries



Note: Retail oil price structures as of January 2016. Price structures in other countries are calculated from benzene prices, whereas in Thailand calculation is based on gasohol 95 price.
Sources: Office of Energy Policy and Planning, and foreign sources. Calculations by Bank of Thailand.

Chart 3 Comparison of headline inflation and long-term inflation expectations



Sources: 1. Inflation expectation surveys by Consensus Economics
2. Inflation expectation models. Calculations based on government bond yields (PIER Discussion Paper No.4)
3. Headline inflation data from Ministry of Commerce

Another option for central banks to enhance the effectiveness of the monetary policy framework to respond to heightened volatility and the evolving global economic landscape is to have **appropriate flexibility for their inflation targets.** The main approach used by most central banks is by setting a **tolerance band** to accommodate various events, especially supply-side factors that may affect the economy and cause inflation to deviate from the target in the short term. For Thailand, the tolerance band for the inflation target is set at ±1.5 percent. The band is slightly broader than ±1.0 percent used in other countries because inflation in Thailand tends to be more volatile, especially during periods of large fluctuations in crude oil prices, because of the smaller proportion of excise taxes in domestic oil prices that caused prices to fluctuate along with global crude oil prices **(Chart 2)**. In addition, retail oil prices account for a large share of Thailand's consumer price basket relative to other countries[2/]. Nevertheless, such a tolerance band is not too wide to affect monetary policy credibility **(Chart 3)**. This was reflected in the latest long-term inflation expectations, which were still close to the inflation target of 2.5 percent.

[2/] In Thailand, oil accounts for 5.5 percent of goods and services in the consumer price basket in 2016, higher than 2.1 percent, 3.4 percent, and 4.5 percent in Japan, Brazil, and the European Union respectively.

Moreover, the transmission of monetary policy takes some time to fully affect the economy and inflation. Attempts to bring headline inflation back to the target within too short a time horizon may increase volatility in the financial markets and jeopardize long-term macro-financial stability. Therefore, **setting a medium-term target** is another approach used by inflation-targeting central banks to look through short-term shocks and allow due consideration for macro-financial stability in the conduct of monetary policy. In this manner, central banks can bring headline inflation to the target in an appropriate timeframe consistent with sustainable economic growth and macro-financial stability. In the case of Thailand, the Monetary Policy Committee (MPC) recognizes the importance of adopting an appropriate inflation-targeting policy horizon. Accordingly, in addition to setting an annual monetary policy target as mandated by the law, the MPC communicates to the public that such annual target also serves as the inflation target for the medium term. From 2016 onward, the memorandum of understanding between the MPC and the Minister of Finance on monetary policy target clearly specified that the inflation target would serve as both the annual target and the medium-term target.

Setting a tolerance band and a medium-term inflation target to enhance the flexibility of the monetary policy framework may cause confusion among the public if inflation breaches the target for an extended period. Therefore, central banks need to **emphasize communication of inflation developments and monetary policy actions** to increase public understanding and anchor the public's inflation expectations. The MPC has duly done so through communication of the medium-term inflation target with a focus on macro-financial stability in monetary policy formulation, including issuing open letters in the case that headline inflation breaches the target.

With regard to 2017, the MPC and the Minister of Finance jointly agreed to set the **headline inflation of 2.5 ± 1.5 percent as the medium-term monetary policy target and the target for 2017**, with cabinet approval on December 20, 2016. As the target is close to those adopted in developing countries that are inflation targeters, it would help maintain Thailand's price competitiveness. Moreover, the tolerance band would help cushion against shocks that may affect inflation and anchor the public's inflation expectations.

The MPC's forecast of the annual average of headline inflation for 2016 at 0.2 percent turned out to be less than the lower bound of the target at 1 percent for two consecutive years. However, the MPC decided to maintain the target adopted for 2015 and 2016 because the Committee anticipated inflation to trend up gradually owing to a variety of factors, and that monetary policy should remain accommodative to support economic recovery. The main reasons that headline inflation stayed below the target were low global crude oil prices and a subdued domestic economic recovery, weighed down by export contractions as the country's trading partners continued to face a slow recovery. However, thanks to some upturn in oil prices since the second quarter of 2016, headline inflation turned positive in April 2016 and edged up gradually. The MPC projected that headline inflation would return to the target within the first quarter of 2017.

As mentioned above, the conduct of monetary policy will become more challenging in the period ahead owing to structural changes in the global economy that may affect the global and domestic inflation dynamics, including **(1) structural changes in global crude oil production** which could prevent sharp rises in oil prices as in the past, **(2) reorientation of the economic structure from manufacturing to services** that could lower inflationary pressures in line with lower service costs stemming from technological developments and innovations, and **(3) aging population** in many countries. The effects of aging on inflation are not yet clear. Inflationary pressures may be lower as an increasing tendency to save could lower consumption.[3/] On the other hand, inflationary pressures may increase as supply cannot keep pace with consumption demand, given production capacity falls following a shrinking labor force.[4/] **The MPC would thus closely monitor these various factors, especially structural factors that could impact inflation dynamics and monetary policy effectiveness, in order to effectively formulate a future monetary policy target and the conduct of monetary policy toward the goals of price stability, economic growth, and macro-financial stability.**

[3/] Anderson, D., D. Botman and B. Hunt (2014) "Is Japan's Population Aging Deflationary?" IMF Working Paper 14/139.

[4/] Juselius, M. and E. Takats (2015) "Can Demographic Affect Inflation and Monetary Policy?" BIS Working Papers No. 485.

Recent Economic Developments

2. Recent Economic Developments

In the third quarter, the global economy recovered at a gradual pace. Advanced economies grew mainly on account of private consumption, consistent with gradual improvements in labor market conditions. Meanwhile, the Chinese economy continued to slow down due to ongoing economic reforms. Meanwhile, Asian economies (excluding China and Japan) grew at a gradual pace as exports slowly picked up. However, growth momentum for the Asian economies in the period ahead might slow down as private investment are affected by increasing uncertainties following the U.S. presidential election.

The Thai economy in the third quarter continued to expand though at a slower pace than the preceding quarter. Tourism was a major growth driver despite a setback from the bombing incidents in the seven southern provinces and the government measure to curb illegal tour operators. Private consumption expanded, supported by both farm and non-farm employment and gradual improvements in farm income after the drought had subsided. The export sector showed signs of improvement in many manufacturing products. Government current expenditure slowed down in the last quarter of the fiscal year after disbursements had been expedited earlier, while capital expenditure continued to expand. However, improvements in private consumption and the gradual recovery of exports were still unable to spur private investment on a broader scale despite growth in investment in certain industries, particularly export-oriented manufacturing.

Headline inflation increased due mainly to higher energy prices. Core inflation remained stable as demand-side inflationary pressures remained subdued. Looking ahead, the Committee expected headline inflation to gradually rise and return to the target band within the first quarter of 2017, though the timing would depend largely on developments in global oil prices.

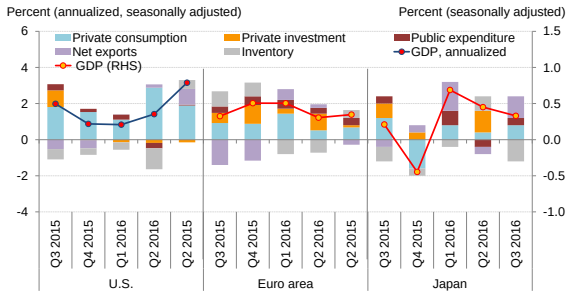
2.1 The global economy

Advanced Economies

Advanced economies slowly recovered on account of private consumption **(Chart 2.1)**. At the same time, monetary conditions tightened after the U.S. presidential election, while the euro and the yen depreciated vis-à-vis the U.S. dollar.

Chart 2.1 Source of growth of G3 economies

(percent change from last quarter)



Source: Bureau of Economic Analysis, Eurostat, Cabinet Office of Japan

The U.S. economy in the third quarter of 2016 grew by 2.3 percent (qoq saar), a rebound from a slowdown during the first half of the year. The expansion was supported by a steady growth in consumption, a smaller contraction of investment in the energy sector following higher oil prices, and an inventory build-up after a consecutive decline in the last five quarters. Recent economic indicators pointed to a continued expansion through increased retail sales, improved confidence, and a steady rise in employment.

In the period ahead, monetary conditions would likely tighten after the Federal Reserve raised its policy rate by 0.25 percent on December 14, 2016. This was the second rate hike since the global financial crisis in 2008 (the first was on December 16, 2015). Moreover, the Federal Reserve's expectations of future rate increases, as reflected in the dot plot, increased from two to three hikes in 2017. Nevertheless, the U.S. economic recovery was expected to continue, driven by consumption that was backed by improved labor market conditions, despite weak private investment. Most analysts viewed that the outcome of the U.S. presidential election would benefit the U.S. economy in the short run due to growth-oriented economic policies, especially the tax deduction which can be implemented rather quickly. However, such policies could affect fiscal sustainability in the future. Meanwhile, inflation was expected to rise and could affect both the Federal Reserve's monetary policy directions and global financial markets conditions going forward (see Chapter 3). Moreover, the impact of trade protectionism policies needs to be monitored (see Box: *The Impact of President-elect Donald Trump's Policy on the Global Economy*).

Euro area economies grew by 0.3 percent (qoq sa) in the third quarter of 2016—a slowdown from the first half of the year during which growth accelerated due to warmer than usual weather. Growth in Germany and Spain was slightly lower than the previous quarter, while France and Italy recorded an improvement. Private consumption, the main growth driver, was supported by gradual improvements in employment and accommodative monetary policy. Recent economic indicators pointed to a gradual recovery of economic activities; in particular, the Purchasing Managers' Index for the manufacturing sector stood above 50.

Looking ahead, the euro area economies were expected to recover slowly with a weaker momentum due to tightened monetary conditions as European government bond yields rose in tandem with U.S. Treasury yields. In the period ahead, the euro area economies would face risks from (1) political uncertainties that could impact the economic recovery, especially the general elections in France and Germany in 2017 and the post-Brexit trade negotiations between the UK and the EU, and (2) risks in the European banking sector. For the latter, although the overall situation gradually improved, some countries, especially Italy, still possessed high ratios of non-performing loans and needed additional capital injection for some banks.

The Japanese economy grew by 0.3 percent (qoq sa) in the third quarter of 2016, down from the first half of the year. Private investment contracted as business sentiment remained low. Meanwhile, government expenditure expanded on the back of spending on repair and reconstruction after the earthquake and additional stimulus measures. Recent economic indicators suggested that economic activities remained on a recovery path as consumption continued to expand given consumer confidence picked up following government stimulus policies.

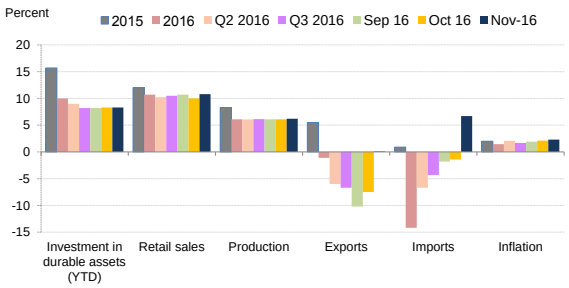
Going forward, the Japanese economy would gradually expand, supported by continued monetary policy easing and government stimulus measures. However, there remained important risks, namely (1) volatility in the global financial market that could impact the yen and (2) transmission of monetary policy and fiscal policy to the real economy that might be less effective than expected.

China

The Chinese economic slowdown continued as a result of ongoing economic reforms to foster long-term stability.

The Chinese economy in the third quarter of 2016 expanded at the same rate as the first half of the year at 6.7 percent. Investment improved in the real estate and manufacturing sectors; infrastructure investment, which remained a key factor supporting China's economic growth, continued to record high growth rates despite some deceleration. Meanwhile, the manufacturing and export sectors continued to slow down. Recent economic indicators suggested a continued expansion from the previous quarter **(Chart 2.2)**. However, infrastructure investment began to slow down after having accelerated in the previous period. Activities in the real estate sector also dampened after the measures were imposed to curb speculation and credit expansion since the end of September 2016.

Chart 2.2 China's economic indicators
(percent change from same month last year)



Source: CEIC

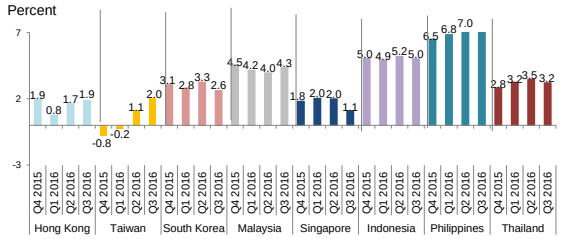
Looking ahead, China's economic slowdown would continue further as a consequence of ongoing economic reforms and stricter monitoring of speculation in the real estate sector. The economy was expected to grow around the official target of 6.5 to 7.0 percent in 2016. Financial stability risks in China, however, continued to warrant close monitoring, namely (1) high level of corporate debt, (2) speculation in the real estate sector, and (3) net capital outflows.

Asia (excluding China and Japan)

Asian economies recovered slowly on the back of gradual improvements in private consumption and exports. However, the recovery for the period ahead might be slower than the previous projection.

Asian economies recovered at a gradual pace in the third quarter of 2016 despite a slowdown in some countries (Chart 2.3). This was partly due to an acceleration of government disbursements during the beginning of a fiscal year. Private investment slowed down due to overcapacity in the manufacturing sector. Moreover, business confidence in some countries, such South Korea and Malaysia, remained weak due to uncertainties surrounding domestic political developments. However, private consumption and net exports continued to expand, especially exports of electronic goods by Taiwan, Hong Kong, and Malaysia, which still benefited from the high-tech cycle upturn thanks to new product launches in the third quarter.

Chart 2.3 GDP growth of Asian economies
(percent change from same quarter last year)



Source: CEIC

Looking ahead, Asian economies would recover gradually but at a slightly slower pace than previously assessed in the previous *Monetary Policy Report*. This could be attributed to an expected slowdown in private investment as a result of increased uncertainty after the U.S. presidential election. Meanwhile, consumption would continue to expand and exports recover on the back of export of electronic goods. However, the economic recovery in the period ahead still faced risks stemming from (1) consumption that might slow down more than expected given the high level of household debt and tighter-than-expected monetary conditions, and (2) structural changes in global trade whereby countries depend more on domestic production, potentially weighing on the export recovery more than expected.

2.2 The domestic economy

The Thai economy in the third quarter continued to expand though at a slower pace compared with the previous quarter **(Chart 2.4).** Tourism was a major growth driver

despite a setback from the bombing incidents in the seven southern provinces and the government measure to curb illegal tour operators. Private consumption expanded, supported by both farm and non-farm employment and gradual improvements in farm income. Exports of many manufacturing products improved. The government's role in driving the economy declined somewhat as government current expenditure slowed down in the last quarter of the fiscal year after disbursements were earlier expedited. However, capital expenditure continued to expand. Nonetheless, improved private consumption and the gradual recovery of exports were still unable to spur private investment on the broader scale despite growth seen in certain industries, especially export-oriented manufacturing.

Chart 2.4 GDP growth[1/]



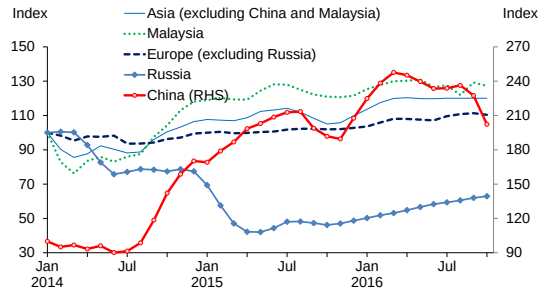
Note: [1/]Calculation based on chain volume measure (CVM)
Source: Office of the National Economic and Social Development Board, seasonally adjusted by Bank of Thailand

Tourism sector remained a major growth driver in the third quarter despite downside risks in the short term from measures to curb illegal tour operators. Nevertheless, these measures would help bring about improvements in the quality of tourism businesses as well as sustainability in longer term.

Tourism continued to record strong growth. Despite the bombing incidents in the seven southern provinces at the beginning of the quarter and measures to curb illegal tour operators at the end of the quarter, the number of Chinese and Malaysian tourists still expanded while European tourist figures also recovered. Moreover, the number of Middle Eastern tourists accelerated after the end of the Hari Raya festival **(Chart 2.5).** Looking ahead, however, tourism faced greater downside risks stemming from the government's measures to curb illegal tour operations. In the short run, such measures would lead to a significant decline in the number of Chinese tourists. On the **supply side**, Thai tour operators delayed or canceled inbound tours from China for fear of being charged. On the **demand side**, the imposition of price floor for tour packages to reflect actual costs made tours to Thailand less cost competitive than countries such as South Korea and Japan. However, survey results indicated that tour business operators in Thailand viewed measures to curb on illegal tour operators to yield medium- to long-term benefits as they improved quality of services. This would in turn improve the overall quality of both operators and tourists which is a key factor that enabled Thai tourism to achieve sustainable growth and greater value-added.

Chart 2.5 Index of foreign tourists classified by nationality

(3-month moving average, seasonally adjusted; January 2013=100)



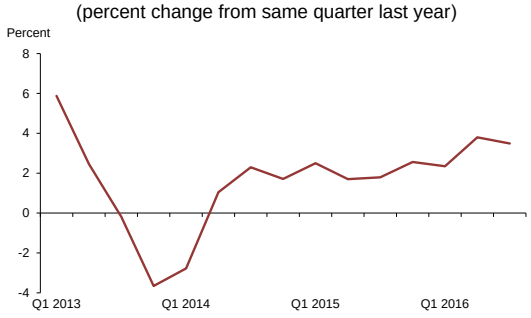
Source: Department of Tourism

Private consumption continued to record a strong expansion, supported by employment, farm income, and government measures.

Private consumption expanded in the third quarter, albeit at a slower rate than the previous quarter **(Chart 2.6).** Supporting factors came from improved employment both in the farm sector after the drought subsided and in the non-farm sectors—both manufacturing and services. Farm income also increased though not for all agricultural products. Moreover, spending on services was stimulated by temporary factors, namely a long holiday in July and the 15,000 baht tax deductions to promote tourism in 2016.

Chart 2.6 Growth of private consumption

(percent change from same quarter last year)



Source: Office of the National Economic and Social Development Board

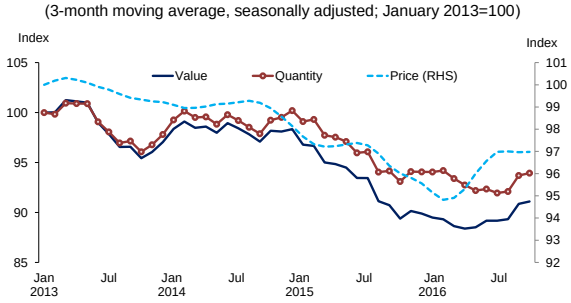
In the period ahead, private consumption would recovery gradually and expected to receive additional push from government stimulus measures, especially the tax deduction of 15,000 baht for domestic spending on goods and services at the end of 2016.

The value of merchandise exports improved in many industries on account of rising demand for some products, expansion into new markets, and partly a result of the consolidation of product location for some products to Thailand such as hard disk drives.

Merchandise exports in the third quarter rebounded in many categories **(Chart 2.7)** thanks to rising demand, expansion into new markets, and the consolidation of product location for some products to Thailand that helped support overall exports. **Exports that expanded on account of rising demand included (1) electronic products and integrated circuits** for (a) smartphones whose demand rose during the launch of new smartphones at the end of the year, (b) electrical appliances with internet connectivity (Internet of Things: IOT), and (c) vehicle parts such as automatic transmission systems and electric vehicles; **(2) automobile and auto parts** due to export to the U.S. and contract manufacturing of pick-up trucks for European carmakers; and **(3) air conditioners** thanks to increased demand from Europe due to the warmer-than-usual weather and rising demand from Thailand's neighboring countries with growing property markets. **Export goods that expanded due to the consolidation of production base to Thailand included**

hard disk drives, although the demand for hard disk drives would likely decline in the long term with the technological shift toward solid-state drives, and **solar cells** whose production base was relocated from China. Nonetheless, the export sector continued to face structural challenges, including declining import dependence on the global scale and Thailand's subdued investment, which together could weaken the recovery of exports in periods ahead.

Chart 2.7 Thai exports (excluding gold) value, price, and volume indices

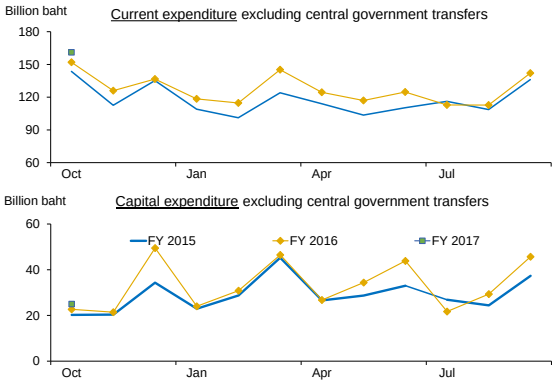


Source: Customs Department, Ministry of Commerce calculations by Bank of Thailand

Public spending continued to drive the economy despite to a lesser extent in the third quarter. While current expenditure slowed down after the acceleration in the previous quarter, capital expenditure continued to expand.

The government's role in driving the economy declined somewhat in the third quarter as current expenditure slowed down in this final quarter for fiscal 2016, partly a result of the acceleration in the previous three quarters. Nonetheless, capital expenditure continued to be well disbursed **(Chart 2.8)** for housing projects, road construction, and irrigation systems. Moreover, capital expenditure of state-owned enterprises continued at a healthy pace, particularly for investment projects according to the infrastructure and electric rail development.

Chart 2.8 Public spending



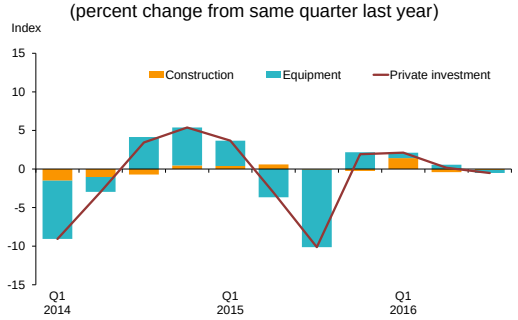
Source: Bureau of Budget; Fiscal Policy Office

Private investment on the whole remained low, despite improvements in some industries with better business and export outlooks.

Private investment remained low, both for investment in equipment and machinery and in construction **(Chart 2.9).** With sufficient production capacity, businesses still awaited clearer trends in global and domestic demand. Investment was mostly undertaken by the same firms in the service sector — those in telecommunication, alternative energy, and logistics and warehouses. Moreover, the improvements in exports had led to more investment in some industries, as reflected by increases in imported capital goods and capacity expansion by export-oriented industries **(Chart 2.10).**

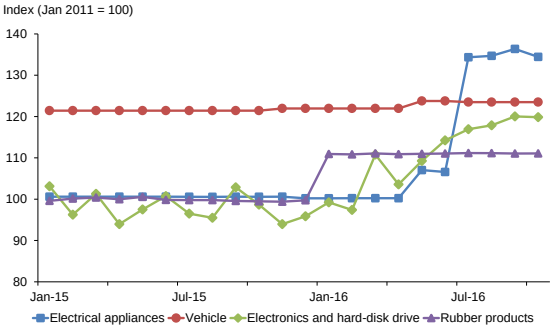
Chart 2.9 Contribution to growth of private investment

(percent change from same quarter last year)



Source: Office of the National Economic and Social Development Board

Chart 2.10 Industrial maximum capacity



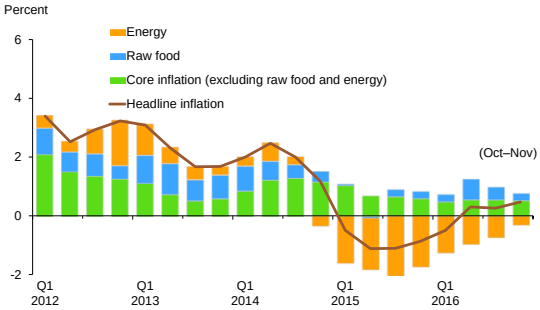
Source: Office of Industrial Economics
Calculations by Bank of Thailand

2.3 Production cost and price conditions

Headline inflation increased mainly due to higher energy prices. Core inflation remained mostly unchanged as demand pressures remained low. In the period ahead, the Committee assessed that headline inflation would rise slowly and return to the target band within the first quarter of 2017, depending largely on developments of prices of crude oil.

Headline inflation increased, averaging at 0.47 percent in the first two months of the fourth quarter **(Chart 2.11).** The increase was due to rising domestic oil prices, as global crude prices increased in anticipation of an agreement to cut production by the Organization of the Petroleum Exporting Countries (OPEC). Meanwhile, prices of fresh food fell on account of lower prices of fruits and vegetables due to increased supply after the drought subsided.

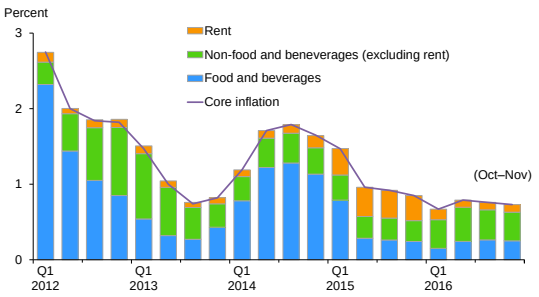
Chart 2.11 Contribution to headline inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce
calculations by Bank of Thailand

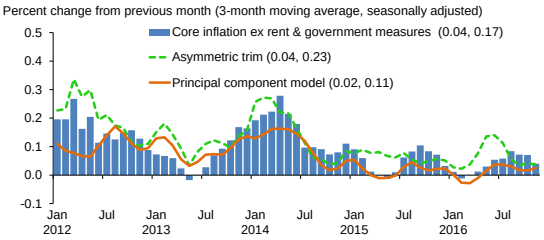
Core inflation averaged at 0.73 percent in the first two months of the fourth quarter **(Chart 2.12**), close to that in the previous quarter. Weak demand-pull pressures were a result of a gradual recovery of the domestic economy, as reflected by underlying inflation indicators which remained at low levels **(Chart 2.13).** Meanwhile, prices of most goods and services in the consumer price basket remained unchanged, indicating limitations of businesses in adjusting prices **(Chart 2.14).**

Chart 2.12 Contribution to core inflation



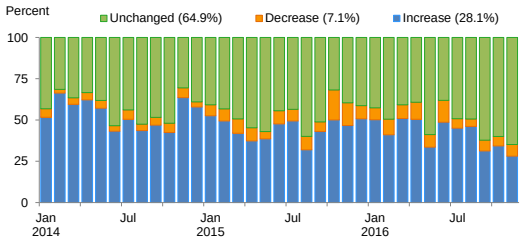
Source: Bureau of Trade and Economic Indices, Ministry of Commerce
calculations by Bank of Thailand

Chart 2.13 Underlying inflation indicators



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of August 2016, while the second value is 2004-2014 average;
Asymmetric trim excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; **Principal component model** calculates changes in common statistical components that attribute price movements across categories of goods and services.
Source: Bureau of Trade and Economic Indices, Ministry of Commerce
calculations by Bank of Thailand

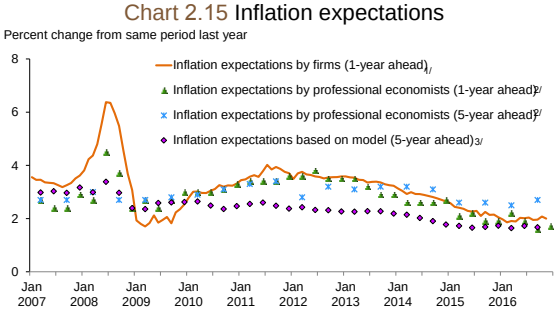
Chart 2.14 Distribution of price changes in the core inflation basket



Note: Calculated from %Mom change in price
Data point indicated in () a proportion of price changes in November 2016
Source: Ministry of Commerce
Calculations by Bank of Thailand

Short-term inflation expectations were down from the previous quarter. One-year-ahead inflation expectations by firms in November 2016 and by professional economists in December 2016 stood at 2.0 percent and 1.7 percent, respectively. Medium-term inflation expectations remained near the inflation target. Five-year-ahead inflation expectations by professional economists stood at 2.5 percent in October 2016 **(Chart 2.15).**

Chart 2.15 Inflation expectations



Source: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

Looking ahead, the Committee expected headline inflation to rise gradually and return to the target band in the first quarter of 2017, supported by increases in crude prices following the cut in oil production by OPEC as well as the low base effect. Furthermore, stronger demand-pull pressures following the economic recovery would allow businesses to raise prices of goods and services. However, uncertainties in global oil prices remained a major risk to inflation forecasts.

Table 2.1 Quarterly inflation

Unit: Percent	2014	2015		2016			
		Q3	Q4	Q1	Q2	Q3	Oct-Nov
Percentage change from previous year (%Δyoy)							
- Headline Consumer Price Index (Headline CPI)	1.89	-1.11	-0.86	-0.50	0.30	0.26	0.47
• Core Consumer Price Index (Core CPI)	1.59	0.92	0.85	0.67	0.79	0.76	0.73
• Raw food	3.46	1.48	1.45	1.52	4.24	2.58	1.47
• Energy	1.68	-16.42	-14.63	-11.41	-8.95	-7.00	-3.03
Percentage change from previous quarter (%Δqoq_sa)							
- Headline Consumer Price Index (Headline CPI)	-	0.0	0.0	-0.3	0.6	0.0	-
• Core Consumer Price Index (Core CPI)	-	0.3	0.2	0.1	0.2	0.2	-
• Raw food	-	1.6	0.3	0.2	2.1	0.0	-
• Energy	-	-4.0	-2.0	-4.7	1.5	-2.0	-

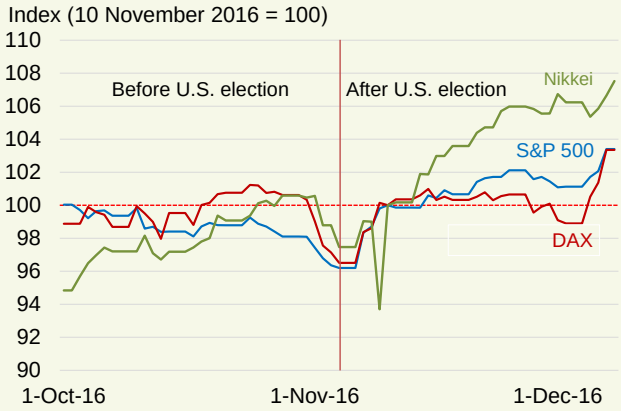
Source: Bureau of Trade and Economic Indices, Ministry of Commerce Calculations of percentage change from last quarter, seasonally adjusted, by Bank of Thailand

The Impact of President-elect Donald Trump's Policy on the Global Economy

The outcome of the U.S. presidential election and policy changes

The outcome of the U.S. presidential election on November 8, 2016 with **Donald Trump's victory was taken positively by investors**, as reflected by stock market gains in major advanced economies **(Chart 1)**. The U.S. economy was expected to benefit in the short term from fiscal stimulus measures, which already raised inflation expectations in major advanced economies.

Chart 1 Stock price indices of major advanced economies



Source: Bloomberg

Major economic policy proposals by President-elect Trump during the election campaign covered several areas. For example, taxation, trade protection, and deportation of immigrant workers **(Table 1).** Clarification on the new U.S. administration's policies in terms of their possibility, scale, and timing of implementation is therefore very important and warrants close monitoring.

Some of the more defined policies were the following: (1) tax policy with the proposed cuts in both personal and corporate taxes could plausibly be implemented around the end of 2017 and the new fiscal year, and **(2) infrastructure investment policy** with private funding through issuance of infrastructure bonds could be only partly implemented, according to analysts' views, because returns on some projects were not very attractive to the investors. However, **other policies remained unclear, such as immigration and trade protection policies.** According to analysts, these policies would be difficult to implement due to several limitations. On immigration, for example, with shortages of the number of public employees, it would take time for the policy to be implemented and need to do it in a gradual pace. Trade protection policy could breach WTO and NAFTA agreements that have limitations from complicated and lengthy legal procedures, with many member countries involved in the process. The Trans-Pacific Partnership (TPP) might be an exception as the U.S. has not officially ratified and could withdraw immediately. However, policies should become clearer after the presidential inauguration in early 2017.

Table1: Donald Trump's election campaign pledges

Policy	Details
Tax reduction	Cut income tax from 39.6 to 33 percent and corporate tax from 35 to 15 percent.
Infrastructure investment	550 billon U.S. dollar infrastructure investment plan to be financed by infrastructure bonds which investors could purchase as well as receive special tax privileges.
Trade	Oppose international trade cooperation with intention to withdraw from the Trans-Pacific Partnership (TPP), renegotiate the North America Free Trade Agreement (NAFTA), and increase tariffs on imports from China and Mexico.
Immigration	Deportation of about 11.3 million unregistered migrants and impose limits on immigration to the U.S.

Source: Compiled by Bank of Thailand

Impact on global economy

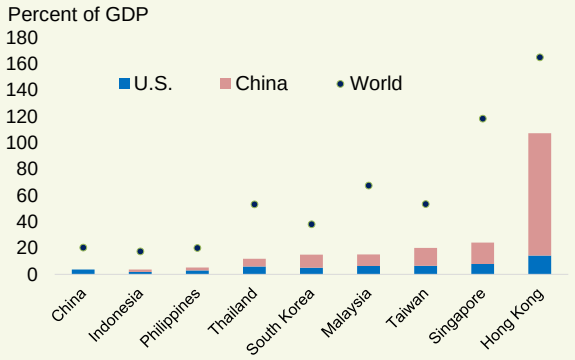
Analysts viewed that the U.S. economy would benefit in the short term from fiscal stimulus measures aimed at boosting household spending and corporate investment even before the official inauguration. Those measures have already shown a positive impact via improved confidence, a stronger U.S. dollar, and higher U.S. Treasury yields following higher inflation expectations. These developments, nonetheless, were expected to also influence the Fed's decisions on policy interest rate rises.

The impact on Asian economies is expected through the following channels.

(1) Trade and investment channel

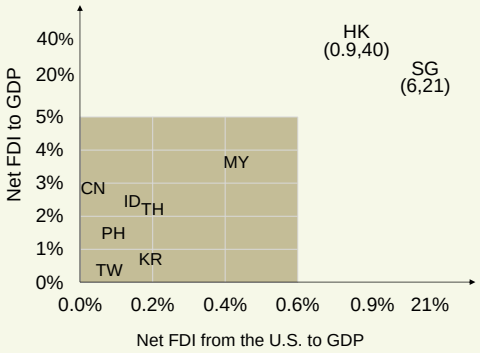
Asian economies would be differently affected depending on their trade linkages and reliance on direct investment from the U.S. While exports of Asian economies could gain from a higher growth of the U.S. economy, especially for countries which have a high level of direct trade with the U.S., the benefit might be reduced due to a more protectionist trade policy. Beside, countries in China's supply chain could be indirectly affected by more U.S. restrictive trade with China **(Chart 2)**. Furthermore, countries that highly depend on direct investment from the U.S. **(Chart 3)** could be affected by the policy supporting U.S. companies to reshore back to the U.S. with the aim to increase employment in the country.

Chart 2 Ratio of exports to GDP of Asian economies to the U.S., China, and the world



Source: Trademap and IMF
Calculation by Bank of Thailand

Chart 3 Investment linkages between Asia and the U.S.



Source: CEIC and Calculations by Bank of Thailand

(2) Financial channel

Volatility in the financial markets heightened following the U.S. presidential election and monetary conditions tightened in some countries as government bond yields in Asia rose in line with U.S. Treasury yields. Moreover, countries with fragile external stability could be facing net capital outflows and rapid currency weakening, which would put constraints on implementing accommodative monetary policy during the economic slowdown.

(3) Confidence channel

Although policies of President-elect Trump remained unclear in many areas, the policy directions were acknowledged and anticipated by investors and the public. The policies were

beginning to impact confidence and investment decisions, especially in countries with new investment project considerations that had to take into account risks from increased policy uncertainty.

In the Committee's assessment, President-elect Trump's policy directions would have a significant impact on the economies and policies of Asian countries including Thailand in the period ahead. Many policies remained unclear in terms of possibility, scale, and timing of implementation and would therefore require close monitoring.

Monetary and Financial Stability

3. Monetary and Financial Stability

Volatility in the global financial markets surged following the U.S. presidential election. U.S. Treasury yields rose with increased inflation expectations. Investors also expected the Federal Reserve to raise its policy interest rate in December. As a result, there were net capital outflows from emerging markets into the U.S. markets. Thailand's overall monetary conditions remained accommodative. Money market interest rates remained at low levels. Meanwhile, corporate costs of funding through the debt market increased from the previous quarter, consistent with the increase in Treasury yields, but remained lower than the average costs in 2015. Relative to the end of last quarter, the baht depreciated against the U.S. dollar, as with most regional currencies, while the nominal effective exchange rate (NEER) strengthened mainly as a result of the yen depreciation against the baht.

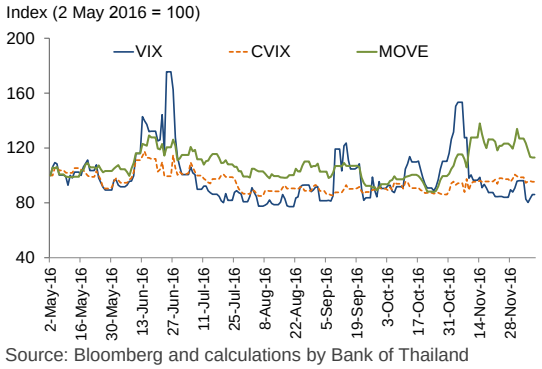
Thailand's overall economic and financial stability remained sound. This was reflected in the high levels of capital buffers and provisions for loan losses maintained by commercial banks to cushion against risks from lower loan portfolio quality. While the search-for-yield behavior continued in the prevailing low interest rate environment, signs of underpricing of risks were limited. Nevertheless, debt serviceability of the household and business sectors, especially SMEs, as well as outstanding supply of residences in certain areas continued to be a concern and warranted continuous monitoring.

3.1 Financial markets[1/]

Global financial markets

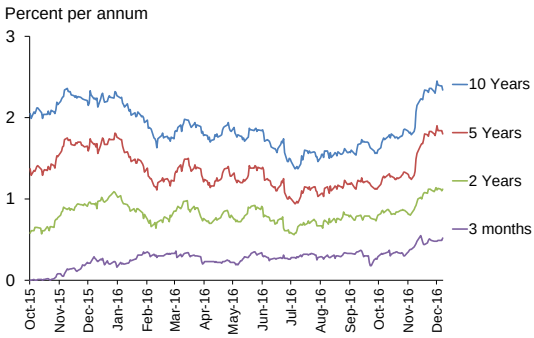
Volatility in global financial markets heightened, as indicated by the volatility indices[2/] **(Chart 3.1)**. In September 2016, volatility temporarily spiked because some investors expected the Federal Reserve to raise its policy interest rate at the policy meeting. In November 2016, higher volatility was observed following Donald Trump's victory in the U.S. presidential election. Expectations that the new U.S. administration's economic stimulation policy such tax reductions and increase in government spending would support a stronger U.S. economic growth and accelerate higher inflation prompted investors to adjust their behavior and inflation expectations. As a result, U.S. Treasury yields rose **(Chart 3.2)**, spurring increases in government bond yields of other countries. In addition, investors expected a higher probability that the Federal Reserve would raise its policy interest rate in December and quicken the pace of interest rate rises in 2017. Due to the releases of the better-than-expected U.S. economic data and the higher inflation expectations, investors withdrew their money from emerging markets (EMs) including Thailand and invested in the U.S. markets instead.

Chart 3.1 Volatility indices



Source: Bloomberg and calculations by Bank of Thailand

Chart 3.2 U.S. Treasury Yields



Source: FRED, Federal Reserve Bank of St. Louis

November 2016 saw the largest capital outflows from EMs since the Taper Tantrum in May 2013. These capital outflows were mostly from bond markets rather than equity markets **(Chart 3.3)**. Most of the outflows were from EMs in the Asia region where they received large capital inflows since the Brexit vote **(Chart 3.4)**. Besides bearing the pressures on gaps of securities return, financial markets in EMs would also

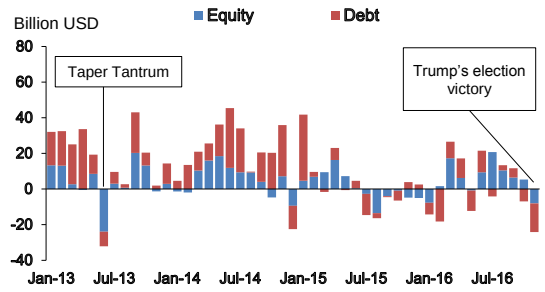
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[1/] Economic data in this Monetary Policy Report are as of December 20, 2016, one day prior to the latest Monetary Policy Committee meeting.

[2/] **VIX** (Volatility Index) is an indicator for stock market volatility, measured by implied volatility of option prices of the S&P500. **Move** (Merrill Lynch Options Volatility Estimate) is an indicator for option bond market volatility, measured by implied volatility of the U.S. government bond market. **CVIX** (Currency Volatility Index) is an indicator for foreign exchange market volatility, measured by implied volatility of 9 major currency pairs.

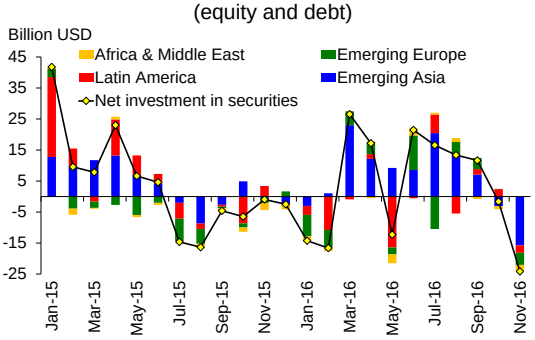
face risks from the implications of the new U.S. government's protectionist trade policy.

Chart 3.3 Capital inflows into assets of emerging economies



Source: Institute of International Finance

Chart 3.4 Net investment in securities (equity and debt)



Source: Institute of International Finance

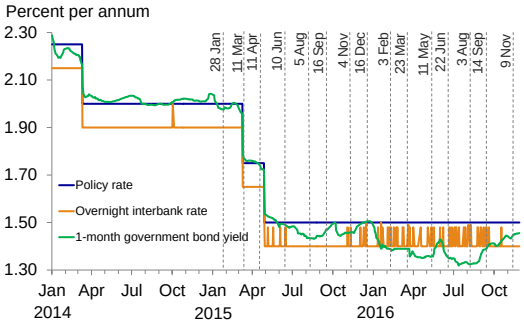
In the period ahead, the MPC would closely monitor developments in global financial markets and international capital flows in view of risks to market volatility which included monetary policy divergence among major advanced economies. The Federal Reserve increased its policy rate on December 14, 2016, in line with investors' expectations. Moreover, the expectations of future rate rises, as indicated in the dot plot, increased from 2 to 3 rate rises in 2017. Meanwhile, the Bank of Japan (BOJ) and the European Central Bank (ECB) continued to adopt a monetary policy easing stance[3/]. There were also geopolitical risks that may affect the economy and financial market volatility, such as the new U.S. government's policies and elections in France and Germany in 2017.

Thai bond and equity markets

Short-term money market rates during October to December 2016 remained close to the policy rate **(Chart 3.5)**. One-month government bond yield increased slightly and moved closer to the policy rate. Long-term government bond yields fluctuated, as influenced by both domestic and overseas factors, before rising along with U.S. Treasury yields following the U.S. presidential election and the policy interest rate hike by the Federal Reserve. As a consequence, long-term government bond yields bounced back to the similar levels as the averages for last year **(Chart 3.6).**

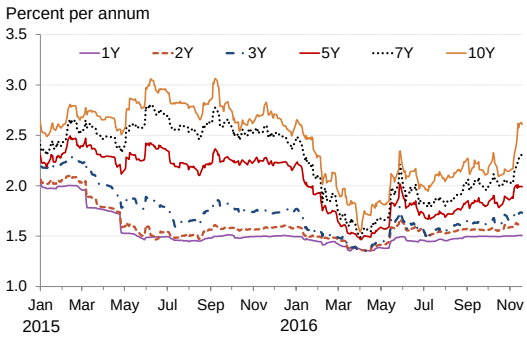
[3/] The BOJ introduced quantitative and qualitative monetary easing with yield curve control with a view to achieving the price stability target of 2 percent (announced on September 21, 2016). The ECB decided to extend its quantitative easing program by nine months until the end of December 2017 changing from the end of March 2017 and scale down its monthly purchases from €80 billion to €60 billion (announced on December 8, 2016).

Chart 3.5 Money market interest rates



Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

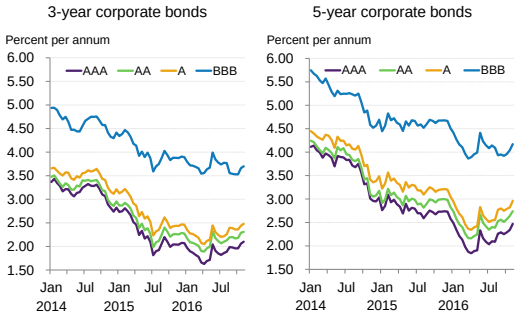
Chart 3.6 Thai government bond yields



Source: Bank of Thailand and Thai Bond Market Association (Thai BMA)

Credit spreads between corporate and government bond yields were largely stable around the previous quarter's level. As a result, costs of financing through corporate bond issuance rose along with the increase in government bond yields which were used as the reference rate. However, overall financing costs stayed below the previous year's averages **(Chart 3.7).**

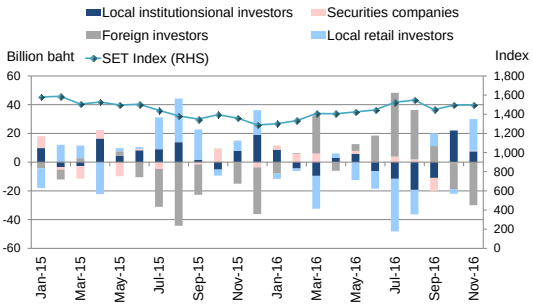
Chart 3.7 Thai corporate bond yields



Source: Thai Bond Market Association (Thai BMA)

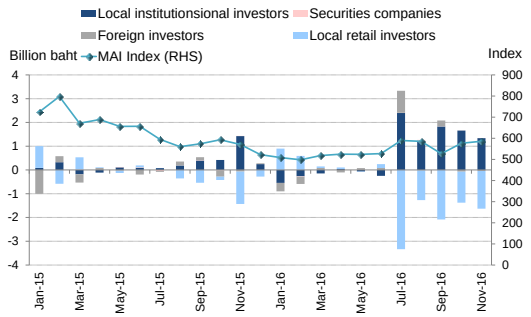
The Stock Exchange of Thailand (SET) Index and the Market for Alternative Investment (mai) Index between October and December 2016 advanced slightly from the previous quarter (Charts 3.8 and 3.9). Local retail and institutional investors were both net buyers, partly as investors priced in the government's stimulus measures and in view of the investment in the energy shares following increases in world oil prices. Meanwhile, foreign investors were net sellers since October and possibly to continue selling until year-end. Foreign investors reduced the weight of Thai and EM assets in their portfolios partly due to the market expectations that the Federal Reserve would raise its policy interest rate in December 2016.

Chart 3.8 Stock Exchange of Thailand Index and net buy classified by investor types



Source: Stock Exchange of Thailand (data as of November 23, 2016)

Chart 3.9 Market for Alternative Investment (MAI) and net buy classified by investor types



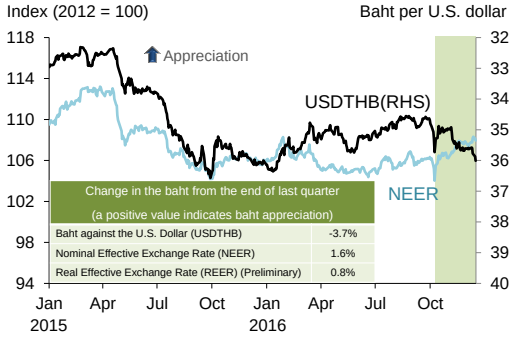
Source: Stock Exchange of Thailand (data as of November 23, 2016)

Foreign exchange market

The baht depreciated against the U.S. dollar relative to the end of the previous quarter, consistent with movements of most regional currencies. Since investors expected a higher probability that the Federal Reserve would raise its policy interest rate in December after releases of better-than-expected U.S. economic figures and higher inflation expectations were shown following the U.S. presidential election, the U.S. dollar was continued to strengthen against bath. As a consequence, foreign investors reduced their holdings in EMs including Thailand. After the Federal Reserve raised its interest rate, the U.S. dollar continued to strengthen in line with market expectations. At times, the baht depreciated more quickly and more steeply than other regional currencies on concerns over the domestic situation. On December 20, 2016, the baht closed at 36.01 baht per U.S. dollar, down 3.7 percent from the end of the previous quarter. The NEER was at 107.96, up by 1.6 percent from the end of last quarter, primarily because of the yen depreciation. Sentiments in global financial markets improved over both economic prospects and policy directions in the U.S., thereby dampening demand for the Japanese yen or considered as safe haven assets. The real effective exchange rate (REER) appreciated by 0.8 percent. The REER's appreciation was smaller than that of NEER because Thai inflation remained lower than other countries **(Charts 3.10 and 3.11).**

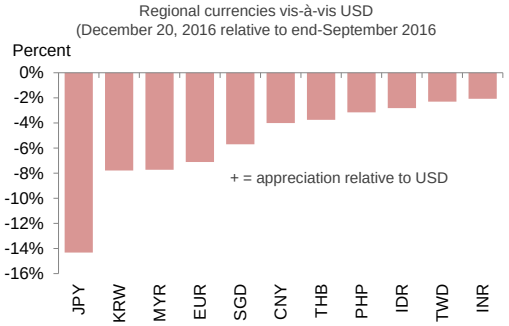
The baht volatility was more pronounced in October, but subsequently the

Chart 3.10 Movements in USDTHB and NEER



Source: Bank of Thailand and Reuters (latest data on December 20, 2016)

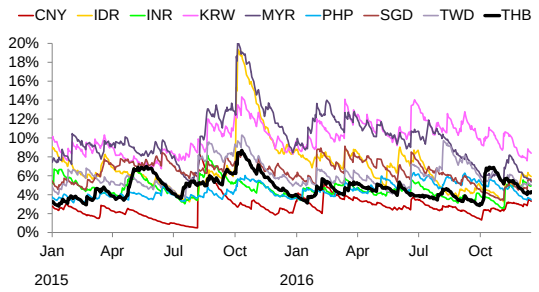
Chart 3.11 Currency movements vis-à-vis USD



Source: Bank of Thailand and Reuters (latest data on December 20, 2016)

baht stabilized after the domestic situation had eased and a clearer picture emerged. Volatility was reduced to a medium level relative to regional currencies **(Chart 3.12)**. Going forward, global capital flows and exchange rates were expected to remain volatile in view of external uncertainties, particularly policy directions of the new U.S. government and monetary policy directions in the advanced economies.

Chart 3.12 Volatility of THB and regional currencies



Note: Volatility calculated by Exponentially Weighted Moving Average (EWMA)
Source: Bank of Thailand and Reuters (latest data on December 20, 2016)

3.2 Financial institutions

Financing costs for firms and deposit rates offered by commercial banks remained steady from the previous report. Meanwhile, overall demand for new financing increased from the second quarter of 2016 across all instruments. This was reflected by an increase in new bond issuances by large corporates and an increase in lending by commercial bank mostly in the form of loans for working capital. Meanwhile, loans for capital investment declined in line with low levels of private investment. Overall credit conditions and outlook showed signs of improvement, although deposit growth reduced banks' needs to mobilize additional deposits as liquidity remained ample.

Interest rates, credits and deposits

Commercial bank lending and deposit rates remained stable overall **(Table 3.1).**

Table 3.1 Benchmark loans and deposit rates of commercial banks*

Percent per annum	2014	2015	2016			
		Q4	Q1	Q2	Q3	Nov 28
12-month deposits						
Average of the 4 largest commercial banks**	1.73	1.40	1.40	1.38	1.38	1.40
Average of remaining banks***	2.22	1.59	1.47	1.42	1.39	1.38
Minimum lending rate (MLR)						
Average of the 4 largest commercial banks	6.75	6.51	6.51	6.26	6.26	6.26
Average of remaining banks	7.44	7.25	7.25	7.16	7.16	7.16
Minimum retail rate						
Average of the 4 largest commercial banks	8.08	7.87	7.87	7.62	7.62	7.62
Average of remaining banks	8.63	8.54	8.54	8.45	8.45	8.45

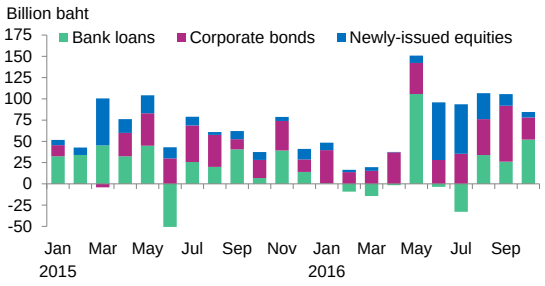
Note: * Benchmark rates averaged across commercial banks at the end of each period

** Four largest commercial banks are BBL, KTB, KBANK and SCB.

*** The following commercial banks are now included under remaining banks: Bank of China (Thailand) (added August 2014), ANZ (Thailand) (added June 2015), and Sumitomo Mitsui Trust Bank (Thailand) (added October 2015). The 12-month average deposit rate of remaining banks in the fourth quarter of 2015 had declined from previous quarter partly because Sumitomo Mitsui Trust Bank (Thailand) offered relatively lower rates.

Total corporate financing between August and October 2016 recorded an increase relative to the second quarter across all instruments (Chart 3.13). The increase was attributed mainly to bond issuances by large corporates as the cost of bond financing remained low. These bond issuers were mostly firms in the wholesale and retail trade, energy, and real estate sectors.

Chart 3.13 Total corporate financing by instruments*

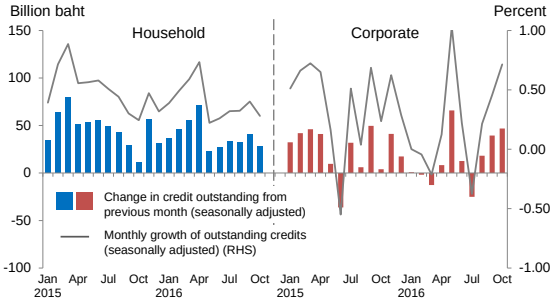


Note: *Monthly change in corporate loans (seasonally adjusted), corporate bonds excluding commercial banks, and newly issued equities.
Source: Bank of Thailand and Thai BMA

Commercial bank lending also increased, especially in new loans to the wholesale and retail trade, telecommunication, and electricity generation and transmission businesses for domestic working capital purposes. Working capital credit for international trade increased somewhat in line with export growth. However, loan demand for investment declined as private investment remained low. New equity issuance were mostly by firms in the wholesale and retail trade, and telecommunication sectors, but the overall level slowed down compared with the previous quarter.

New loans to households[4/] between August and October 2016 increased at a slower pace relative to the second quarter (Chart 3.14). The slowdown was pronounced in new loans from specialized financial institutions after the government's real estate stimulus measures had ended. Meanwhile, new loans from commercial banks increased, particularly credit card and car loans. The slowdown in new loans extended to households was consistent with the findings from the Credit Conditions Survey in the third quarter of 2016. The Survey indicated that loan demand from households was lower than in the previous quarter as consumer sentiment remained soft while financial institutions were also more cautious over car loans and mortgage loans.

Chart 3.14 New private credits



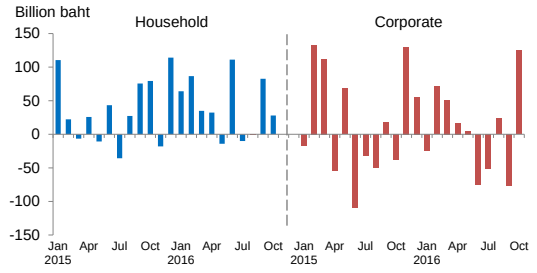
Note: New corporate credits calculated from seasonally adjusted change in credits outstanding of Other Depository Corporations (ODCs)
Source: Bank of Thailand

Deposit outstanding at the end of October 2016 increased from the end of July, especially household deposits (Chart 3.15). Growth of deposits including bills of exchange (B/E) slightly accelerated. Credit

4/ Calculated from change in outstanding credit (seasonally adjusted) of other depository corporations (ODCs). ODCs include commercial banks, specialized financial institutions that are banks owned by the government, finance companies, saving cooperatives, and mutual funds.

conditions and trends showed signs of improvement, but special deposit rates offered by commercial banks were unchanged from the previous quarter **(Chart 3.16)** as there was no need for banks to mobilize additional deposits given stable bank liquidity. This was reflected in the **lower loan-to-deposit (including B/E) ratio relative to the second quarter of 2016** as deposit growth outpaced loan growth **(Chart 3.17).**

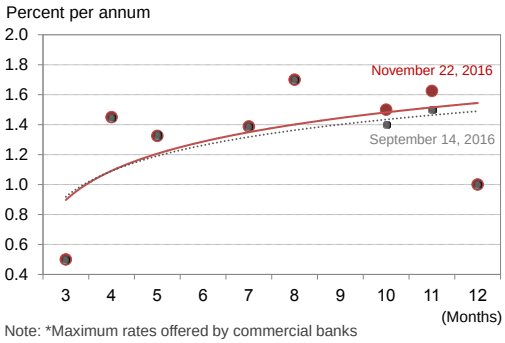
Chart 3.15 New deposits*



Note: * Change in outstanding deposits at depository institutions (excluding Bank of Thailand) from the previous month. The amounts only account for new deposits, not including transfers within and between commercial banks or rolled-over deposits.

Source: Bank of Thailand

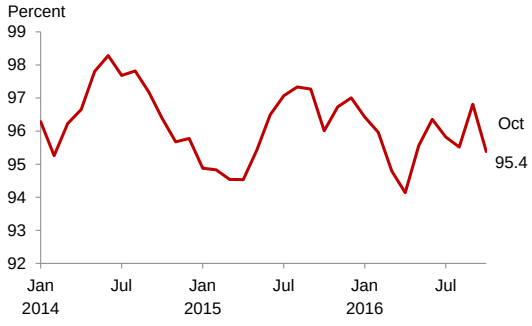
Chart 3.16 Special deposit rates*



Note: *Maximum rates offered by commercial banks

Source: Bank of Thailand

Chart 3.17 Loan to deposit (including B/E) ratio of commercial banks



Source: Bank of THailand

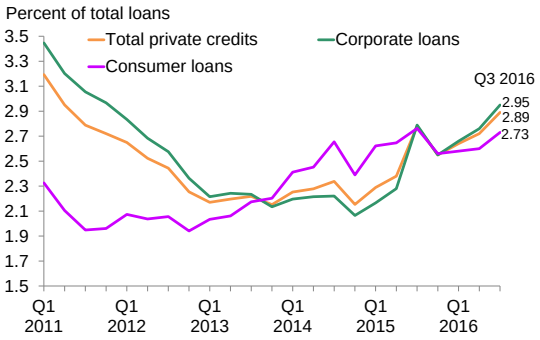
Financial stability

Financial stability remained sound overall, despite deterioration in credit quality of consumer and business loan due to the extend period of the slow economic recovery. Nevertheless financial institutions stability remained strong as banks continued to closely monitor credit quality and maintained sufficiently high levels of capital buffers and loan-loss provisions to cushion against risks stemming from declining loan quality. Meanwhile the search-for-yield behavior remained at large but posed limited risks to the financial system as a whole.

The credit quality of commercial bank loans at the end of the third quarter declined from the previous quarter due to the gradual economic recovery. This was reflected in the ratio of non-performing loans to total loans (NPL ratio) which rose from 2.72 percent at the end of the second quarter to 2.89 percent, attributable to the increase in NPL ratios of among both business loans (especially loans to SMEs) and consumer loans **(Chart 3.18).** The NPL ratio of SMEs continued to rise from 3.77 percent at the end of the second quarter to 4.04 percent, attributable to deterioration in credit quality in the manufacturing, wholesale and retail trade, and construction sectors given the gradual economic recovery. At the same time, the NPL ratio among large corporates

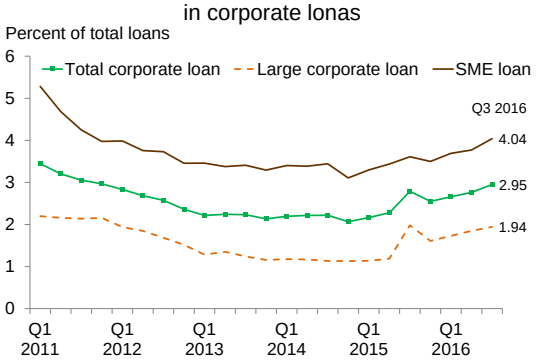
rose from 1.85 percent in the previous quarter to 1.94 percent **(Chart 3.19)**.

Chart 3.18 Non-performing loans of commercial banks



Source: Bank of Thailand

Chart 3.19 Share of non-performing loans in corporate lonas



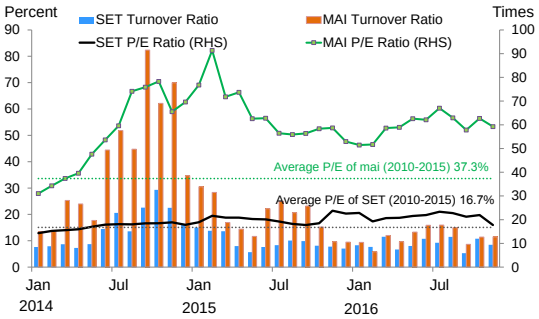
Source: Bank of Thailand

Although the deterioration in credit quality of commercial bank loans affected profitability, but financial institutions stability remained strong. Commercial banks continued to closely monitor credit quality and maintained sufficiently high levels of capital buffers and loan-loss provisions to cushion against risks stemming declining loan quality. The actual to regulatory loan loss provision ratio in the third quarter of 2016 was at 164.4 percent while the capital adequacy ratio was at 18.5 percent, which is higher than the minimum set by the BOT at 8.5 percent.

The search-for-yield behavior remained at large amidst the prolonged low interest rate environment.

With regard to **equity market** movements, the SET Index continued on an upward momentum, and stock prices were high as reflected by the **price-earnings ratios (P/E ratio) for the SET and the mai which were above their historical averages (Chart 3.20).** Nevertheless, signs of market overheating were not present since the turnover ratios of both markets remained low. Additionally, transactions were mainly settled in cash, and margin loans remained low at 12 percent of total transaction value as of October 2016.

Chart 3.20 Turnover ratio and Price-Earnings Ratio (P/E)* in SET and MAI

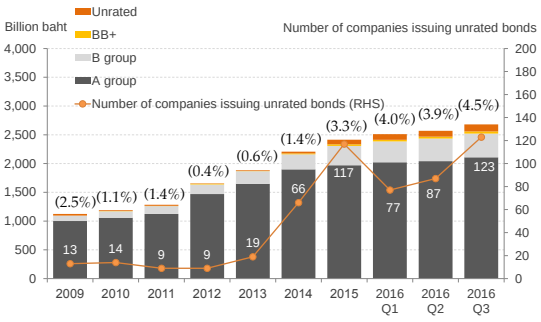


Note: * Current price to Earnings Per Share (EPS) over the last 12 months
Sources: Stock Exchange of Thailand and Bloomberg
(data as of November 23, 2016)

As for activities in the **bond market,** corporate bond issuance continued to rise and although most of these issuance possessed good credit ratings, issuance of unrated bonds by unrated companies were also on the rise. The proportion of unrated bonds to outstanding corporate bonds rose from 3.9 percent in the previous quarter to 4.5 percent in the third quarter **(Chart 3.21)**. A large share of unrated bonds were of shorter

maturities at 42 percent. The presence of roll-over risks among these issuance might pose concerns with regard to liquidity risks. **Nevertheless, systemic risks were deemed to be rather limited** as unrated bonds were mainly sold to institutional investors or accredited investors.

Chart 3.21 Corporate bonds outstanding (short and long maturities) classified by credit rating



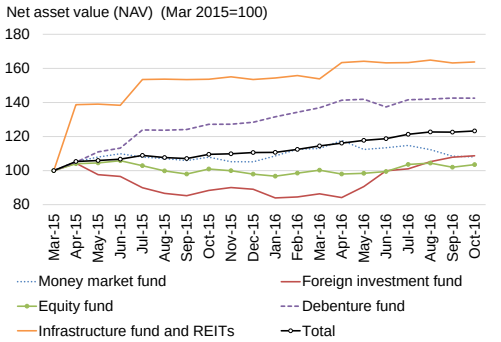
Note: The A group consists of AAA to A-, while the B group consists of BBB+, BBB, and BBB-

* Percent of non-rated bonds in total corporate bonds

Source: Thai Bond Market Association (Thai BMA)

Mutual funds continued to grow as indicated by the increase in the net asset value (NAV) from early September to October 2016, especially of foreign investment funds (FIFs) **(Chart 3.22)** which mainly focused on investment-grade securities in the United Arab Emirates, the U.S., Luxemburg, Hong Kong, and Qatar. Funds that possessed deposit-like characteristics remained popular, with growth observed in daily fixed income funds (daily FI), while money market funds (MMFs) declined in part due to investors' search for higher yields. Although daily FI faced less stringent regulations and were able to invest in riskier debt securities than MMFs, risks remained contained as their investments focused on investment-grade bonds with 95 percent of their holdings in public-sector bonds, deposits and short-term debt instruments with A credit rating, while 31 percent of their total assets were considered highly liquid assets. In addition, the Office of the Securities Exchange Commission (SEC) continuously monitored bond issuance activities and oversaw that issuers appropriately disclose information to ensure that investors have proper understanding of what they are investing in, and tightened bond issuance standards.

Chart 3.22 Investment in mutual funds



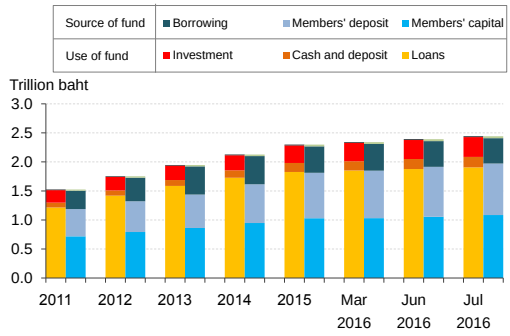
Note: Excluding Vayupak Fund 1, Country Fund, and Real Estate Fund 2-4

Source: Association of Management Companies

Calculations by Bank of Thailand

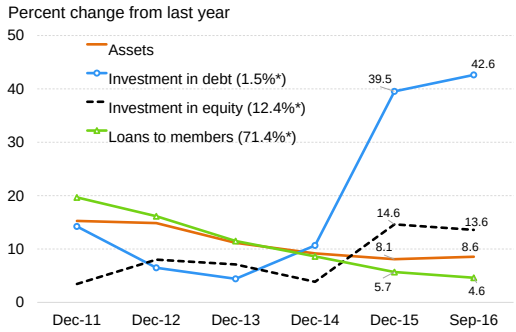
Savings cooperatives' deposits and shares also gained traction among savers as they offered higher interest rates and dividend yields than bank deposit rates **(Chart 3.23)**. The search-for-yield behavior was also observed in some savings cooperatives, as evident in their increasing proportion of equity assets while loans to members grew only slightly. However, equity assets still made up a small 1.5 percent of their total assets **(Chart 3.24).**

Chart 3.23 Saving cooperatives' use and source of fund



Source: Cooperative Auditing Department, calculations by Bank of Thailand

Chart 3.24 Growth in asset and investment of saving cooperatives



Note: *Ratio to total assets of saving cooperatives in September 2016

Source: Cooperative Auditing Department, calculations by Bank of Thailand

Overall, the search-for-yield behavior remained at large as interest rates are expected to remain low for a longer period of time, but the risks it posed to the financial system as a whole remained limited and signs of broad-based underpricing of risks were not present. Nonetheless, the Committee would continue to monitor such behavior closely, especially among investors who continued to seek higher returns, and mutual funds and savings cooperatives that tended to invest in riskier assets, as well as competition among financial institutions in extending credit to large corporates and only certain sectors.

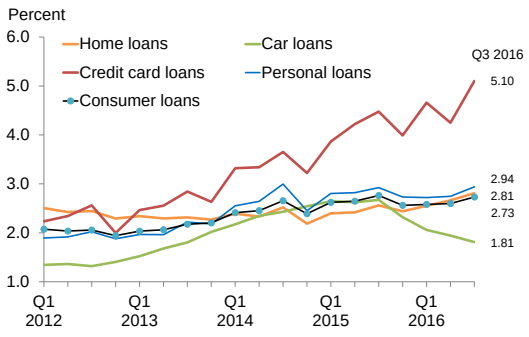
3.3 Non-financial sectors

Household sector

Households' debt serviceability continued to decline, as reflected in the increase in NPL ratios of most types of consumer loans. Meanwhile, the slow economic recovery and high debt levels limited households' ability to adjust financially. However, slower growth of household debt and improved farm income as well as maturing first-car loan debts would improve the financial position of the household sector going forward.

Households' debt serviceability continued to decline, as reflected in the increase in NPL ratios of almost all types of consumer loans, except car loans, from 2.60 percent in the previous quarter to 2.73 percent in the third quarter **(Chart 3.25)**. Moreover, some households, especially those belonging to the low and medium income groups, may face intensified liquidity problems amid the slow economic recovery, as reflected by their increasing reliance on consumer loans from non-banks, a concern which would be monitored closely going forward.

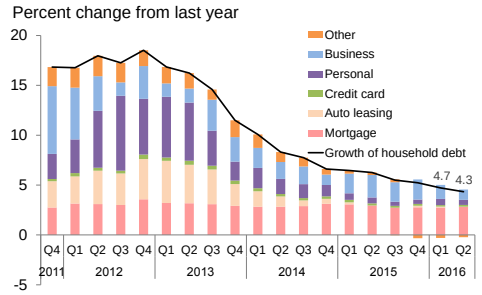
Chart 3.25 Share of non-performing loans in consumer loans



Source: Bank of Thailand

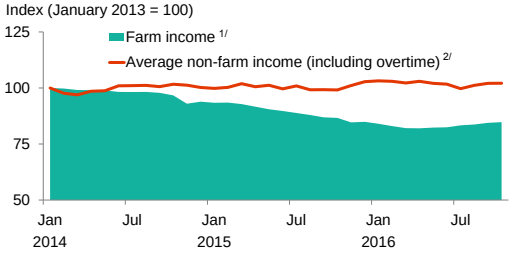
Although household debts was expected to decelerate (Chart 3.26), the ratio of household debt to GDP remained at a high 81.3 percent in the second quarter, edging down from 81.5 percent in the previous quarter, thus constraining household ability to adjust financially going forward. Household income, especially related to the service sector, may be affected by the measures to curb illegal tour operators. Nevertheless, certain positive factors were expected to boost household's debt serviceability such as (1) the uptrend in income of farm households **(Chart 3.27)**, thanks to the government's financial assistance for the agricultural sector and the positive outlook for global prices of some commodities, (2) maturing debt from the first-car loans, and (3) minimum wage increases effective from the beginning of 2017.

Chart 3.26 Contribution to growth in household debt by purpose[1/]



Note: [1/] Loans to households from financial institutions
Source: Bank of Thailand

Chart 3.27 Household income



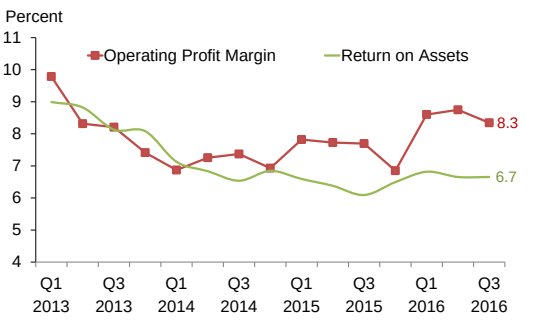
Note: [1/] Seasonally adjusted, 12-month moving average
[2/] Seasonally adjusted, 3-month moving average
Sources: Office of Agricultural Economics, National Statistical Office; calculations by Bank of Thailand

Corporate sector

Performance of non-financial companies listed on the SET declined slightly in the third quarter of 2016. Debt serviceability also declined, especially for SMEs whose financial positions were vulnerable to the slow economic recovery.

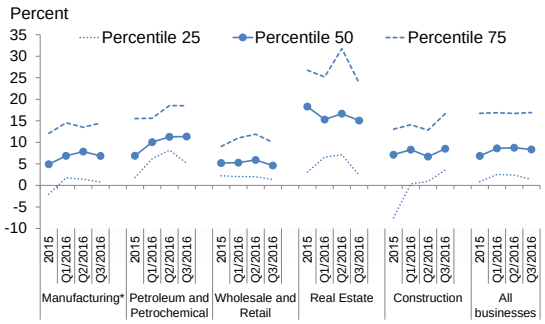
Overall business performance in the third quarter remained sound despite a decline from the previous quarter. Corporates' revenue and profitability weakened somewhat, as reflected in a lower operating profit margin and return on assets **(Chart 3.28)** due to weaker sales figure after government's stimulus measures for the service and the real estate sectors ended. Meanwhile, sales in some sectors such as the construction sector remained robust due to progress in the public- sector infrastructure projects. Amid the gradual economic recovery, most businesses focused on cost management and other controllable factors, and thus were able to maintain high operating profit margins compared to the previous year **(Chart 3.29).**

Chart 3.28 Operating profit margin and return on assets



Note: Median estimates
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

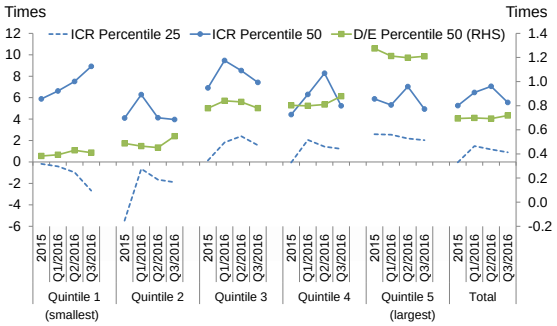
Chart 3.29 Operating profit market by sector



Note: *Exclude Petroleum and Petrochemical

Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Chart 3.30 Interest Coverage Ratio (ICR)* and Debt to Equity (D/E) ratios classified by firm size



Note: *The 25th percentile reflects firms with low debt serviceability; ICR calculation is annualized; D/E is median estimate; Firms divided based on their asset size into 5 quintiles from smallest to largest.

Overall corporate debt stabilized at a level close to the previous quarter. The median D/E ratio remained low at 0.7. Overall **debt serviceability declined somewhat due with lower profitability but remained at a healthy level** with the median ICR at 5.8, well above the international standard of 1.5 **(Chart 3.30)**. Although most business sectors were able to make adjustments amid the gradual economic recovery and low interest costs, SMEs still had limited room to make adjustments and their financial positions remained fragile. Classification of listed companies into five quintiles based on asset size revealed that some companies in the first two quintiles continuously experienced losses, as reflected in the continuously negative ICR (in first and second quintiles) at the 25th percentile. This indicates increasing debt default risks of small companies. Using these smaller companies as representatives for non-listed SMEs, it could be inferred that debt serviceability of SMEs deteriorated. This was consistent with the NPL ratio of SMEs at the end of the third quarter of 2016 which was expected to rise, especially in the manufacturing and wholesale and retail trade sectors.

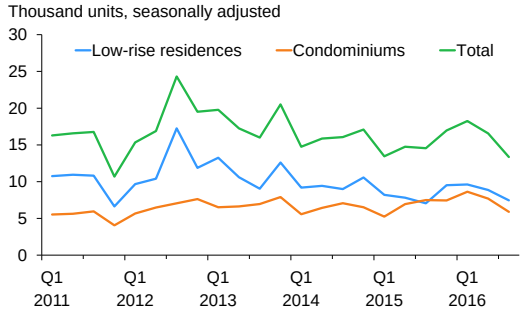
Real estate sector

The property market in Bangkok and its vicinities continued to slow down in the third quarter of 2016 after the government's stimulus measures for the sector had ended in the beginning of the year. Going forward, risks of unsold condominium units in some areas and price ranges, and lower debt serviceability of some operators that had weaker financial positions continued to warrant close monitoring.

The property market in Bangkok and its vicinities continued to slow down in the third quarter of 2016 as **demand** receded after the reduced fee on ownership transfer ended in April 2016. Commercial banks also adopted more stringent lending standards given that loan affordability of buyers declined with prevailing high debt burden. This was reflected in the smaller number of banks' new housing loan approvals in Bangkok and its vicinities **(Chart 3.31)**. Demand was expected to slow down in the fourth quarter due to the mourning period, which was conducive to property transactions. Post-finance quality also dropped, as reflected in the upward NPL ratio of

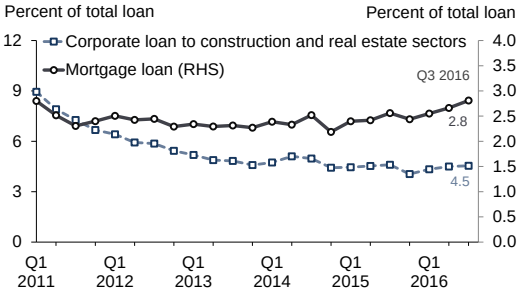
housing loans in the third quarter of 2016 **(Chart 3.32)**. This might prompt banks to adopt more stringent lending standards.

Chart 3.31 Residential units in Bangkok and its vicinity with approved mortgages by commercial banks



Source: Bank of Thailand

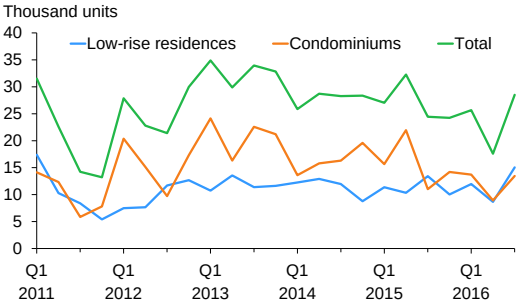
Chart 3.32 Quality of mortgage loans classified by borrowers



Source: Bank of Thailand

On the **supply** side, the third quarter saw an increase in the number of new projects launched in Bangkok and its vicinities **(Chart 3.33)**. However, with fewer new projects launched in the first half of 2016, the total number of new projects launched in the whole year was lower compared to 2-3 years ago.

Chart 3.33 New residential projects launched in Bangkok and its vicinity

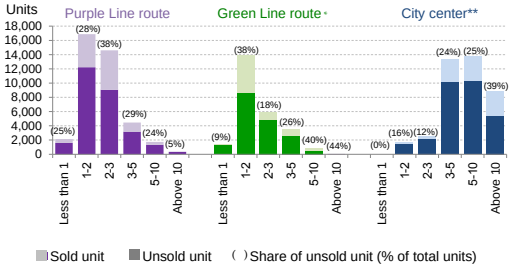


Source: Agency for Real Estate Affairs (AREA), calculations by Bank of Thailand

Risks to monitor in the period ahead would include

(1) While outstanding supply does not pose a major concern as the absorption rate of housing projects and condominium projects in 2016 stabilized at low levels at 2.4 and 1.3 years respectively, the likely increase in outstanding supply in certain areas and price ranges continued to warrant monitoring, for instance condominium units costing 2-5 million baht along the Purple Line route and units costing over 5 million baht along the Green Line extension route (Bearing-Samut Prakarn) **(Chart 3.34).**

Chart 3.34 Share of unsold condominium units in the first half of 2016



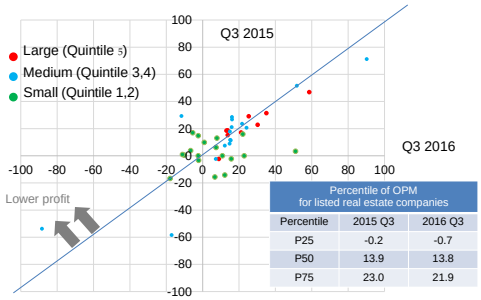
Note: * Bearing-Samutprakarn route
** Covering areas in Phaholyothin, Pathumwan, Sukhumvit, Rama 4, Yannawa, Silom and Phranakorn

Source: AREA and calculations by Bank of Thailand

(2) Adverse effects of the demand slowdown on debt serviceability of small and medium sized developers. Classification of listed property developers into five quintiles

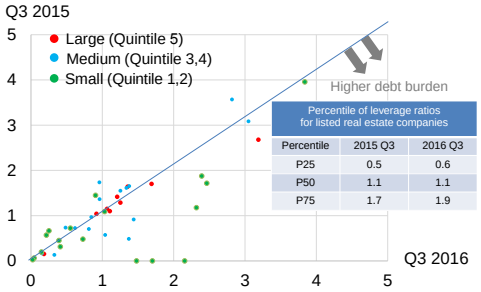
based on asset size reveals that small and medium sized developers experienced lower profitability while their debt burden increased, as indicated in shrinking operating profit margins **(Chart 3.35)** and higher D/E ratios **(Chart 3.36).**

Chart 3.35 Operating Profit Margin of real estate developers (Q3 2016 compared with Q3 2015)



Note: Listed real estate developers are classified into 5 quintiles according to asset size
Source: Stock Exchange of Thailand

Chart 3.36 Debt to Equity ratio (D/E) of real estate developers (Q3 2016 compared with Q3 2015)

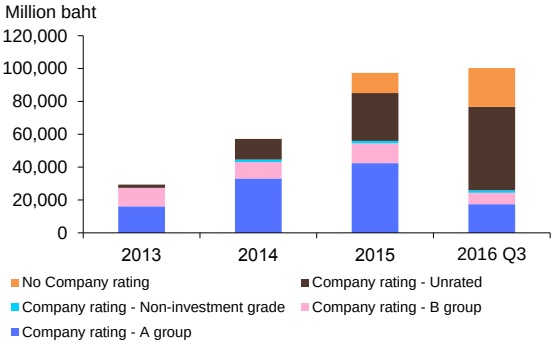


Note: Listed real estate developers are classified into 5 quintiles according to asset size
Source: Stock Exchange of Thailand

In addition, in recent periods, developers became less reliant on borrowing from banks and shifted to alternative funding sources such as bond issuance, in particular unrated bonds **(Chart 3.37)**.

Hence, monitoring of possible systemic risk would be needed going forward in the event of debt default by issuers of unrated bonds, which may have repercussions on other operators via a fall in bond market investors' sentiment.

Chart 3.37 Outstanding of new debt issuances by real estate developers (short- and long-term)



Source: ThaiBMA (only listed debentures)

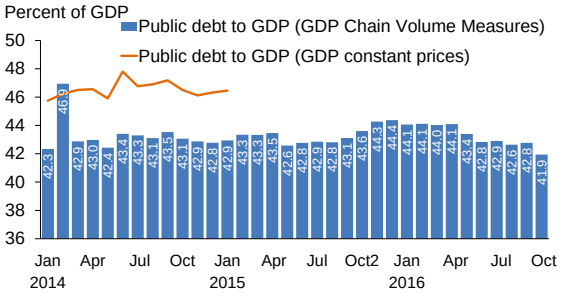
Fiscal stability

Fiscal stability remained sound, as reflected in the public debt level which stayed below the fiscal sustainability threshold and the treasury cash balance that was deemed sufficient to meet unexpected fiscal needs. Nonetheless, pressures on fiscal stability would likely increase in the periods ahead from additional borrowings needed for investment in transportation infrastructure projects, as well as future fiscal burden arising from recently approved stimulus measures.

Fiscal stability remained sound, as reflected in the public debt to GDP ratio which remained below the fiscal sustainability threshold of 60 percent. As of October 2016, the ratio stood at 42.7 percent, slightly up from 42.6 percent in August 2016 **(Chart 3.38)**, mainly due to principal repayment
to the Bank for Agriculture and Agricultural Cooperatives (BAAC) and bond redemption by PTT Plc. Treasury balance stood at 230 billion baht at the end of October 2016, down from 303 billion baht at the end of August 2016, primarily due to disbursements of the regular budget, which mainly comprised of

subsidies to the Department of Local Administration and the National Health Security Fund as well as compensation of principal and interest repayments on the Rice Pledging Scheme to the BAAC.

Chart 3.38 Ratio of public debt to GDP



Note: (1) Chart shows calendar years, (2) Official figures for debt-to-GDP ratio based on GDP chain volume measures are available only for February and March 2015. Figures prior to this period are calculated by the Bank of Thailand

Source: Public Debt Management Office

Going forward, pressures on fiscal stability are likely to increase due to (1) borrowing for investment in infrastructure projects according to the transport infrastructure development strategy 2015-2022, especially rail and road projects, will largely raise public debt in the next 2-3 years, and (2) additional fiscal burden from quasi-fiscal activities that were part of the economic stimulus packages including compensation for the reduction of interest income to specialized financial institutions taking part in the soft loan program. In addition, if the government continued to implement tax reduction measures, government revenue would likely fall and pose risks to fiscal stability.

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2015	2015			2016				
			H1	Q3	Q4	Q1	Q2	Q3	Oct	Nov
1. Financial market sector										
Bond market										
Bond spread										
(10 years - 2 years)	1.3	1.1	0.9	1.2	1.1	0.7	0.5	0.6	0.5	1.1
Equity										
SET index (End of period)	1,497.7	1,288.0	1,504.6	1,349.0	1,288.0	1,407.7	1,445.0	1,483.2	1,495.7	1,510.2
Actual volatility (SET Index)[1/]	12.5	13.9	12.0	12.3	13.7	15.9	11.9	13.6	23.7	10.8
Price to Earnings Ratio (เท่า)	17.8	22.6	20.1	17.8	22.6	20.7	22.0	21.3	22.0	18.0
FX market										
Actual volatility (บาท) (%annualize)[2/]	4.0	5.1	4.5	5.2	6.1	4.4	4.6	3.8	5.4	5.5
Nominal Effective Exchange Rate (NEER)	104.3	108.5	110.8	106.6	105.9	106.7	105.1	105.8	106.0	107.1
Real Effective Exchange Rate (REER)	103.0	104.4	106.99	102.4	101.2	100.8	100.0	100.3	100.3	N.A.
2. Financial institutions sector [3/]										
Minimum lending rate (MLR)[4/]	6.75	6.5	6.51	6.50	6.50	6.51	6.26	6.26	6.26	6.26
12-month fixed deposit rate[4/]	1.73	1.40	1.50	1.46	1.40	1.40	1.38	1.38	1.40	1.40
Capital adequacy										
Regulatory capital to risk-weighted asset (%)	16.8	17.4	16.8	17.3	17.4	17.5	17.5	18.5	N.A.	N.A.
Earning and profitability										
Net profit (billion baht)	214.1	192.3	103.7	50.0	38.6	51.2	50.9	49.8	N.A.	N.A.
Return on assets (ROA)	1.2	0.9	1.3	1.2	0.9	1.2	1.2	1.1	N.A.	N.A.
Liquidity										
Loan to deposit and B/E	95.7	97.0	96.5	97.3	97.0	94.8	96.4	96.9	N.A.	N.A.
3. Household sector										
Household debt to GDP (times)	79.9	81.6	80.6	80.8	81.6	81.5	81.3	N.A.	N.A.	N.A.
Financial assets to debt (times)	2.5	2.6	2.5	2.5	2.6	2.5	2.5	N.A.	N.A.	N.A.
Proportion of loans that are 3 months delinquent (%)										
Thai commercial banks :										
Consumer loans	2.4	2.6	2.6	2.8	2.6	2.6	2.6	2.7	N.A.	N.A.
Housing loans	2.2	2.4	2.4	2.6	2.4	2.6	2.7	2.8	N.A.	N.A.
Auto leasing	2.5	2.3	2.6	2.7	2.3	2.1	1.9	1.8	N.A.	N.A.
Credit cards	3.2	4.0	4.2	4.5	4.0	4.7	4.2	5.1	N.A.	N.A.
Other personal loans	2.5	2.7	2.8	2.9	2.7	2.7	2.7	2.9	N.A.	N.A.
4. Non-financial corporate sector[5/]										
Operating profit margin (%)	7.1	7.5	7.7	7.7	6.8	8.6	8.7	8.3	N.A.	N.A.
Debt to equity ratio (times)	0.8	0.7	0.7	0.7	0.7	0.7	0.7	0.7	N.A.	N.A.
Income coverage ratio (times)	5.8	5.5	5.8	5.4	5.3	6.5	7.1	5.5	N.A.	N.A.
Current ratio (times)	1.6	1.7	1.7	1.6	1.7	1.7	1.7	1.7	N.A.	N.A.

Table 3.2 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2014	2015	2015			2016				
			H1	Q3	Q4	Q1	Q2	Q3	Oct	Nov
4. Non-financial corporate sector5/ (Continues)										
Delinquency and NPL ratio										
Large business	1.1	1.6	2.0	2.0	1.6	1.7	1.9	1.9	N.A.	N.A.
SMEs	3.1	3.5	3.4	3.6	3.5	3.7	3.8	4.0	N.A.	N.A.
5. Real estate sector										
Demand										
The number of approved mortgages from banks										
(Bangkok and its vicintly)	62,839	59,667	26,503	15,770	17,394	15,487	15,611	14,149	4,596	5,301
Single-detached and semi-detached houses	15,694	13,152	6,384	3,115	3,653	3,511	3,503	3,216	758	1,012
Townhouses and commercial buildings	21,764	19,210	9,199	4,812	5,199	5,092	5,410	4,718	1,344	1,593
Condominiums	25,381	27,305	10,920	7,843	8,542	6,884	6,698	6,215	2,494	2,696
Supply										
The number of new openings (Bangkok and its vicintly)	111,211	107,988	59,316	24,436	24,236	25,667	17,584	28,515	15,397	8,339
Single-detached and semi-detached houses	18,933	17,637	6,354	7,388	3,895	4,655	2,918	5,379	1,330	1,697
Townhouses and commercial buildings	26,980	27,518	15,349	6,034	6,135	7,308	5,735	9,674	2,807	1,152
Condominiums	65,298	62,833	37,613	11,014	14,206	13,704	8,931	13,462	11,260	5,490
Housing price index[5/]										
Single-detached houses (including land)	127.9	129.3	127.7	129.6	129.3	129.4	129.4	131.3	128.6	129.3
Townhouses (including land)	133.5	137.5	133.7	134.7	137.5	138.8	138.8	136.8	135.6	135.3
Condominiums	151.1	160.9	152.1	156.4	160.9	160.0	160.0	169.2	172.9	172.7
Land	145.6	168.8	151.9	157.8	168.8	169.5	169.5	170.2	170.0	170.2
6. Fiscal sector										
Public debt to GDP (%)	42.8	44.4	42.7	43.1	44.4	44.0	42.8	42.8	42.7	N.A.

[1/] Annualized standard deviation of return

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 4 largest Thai commercial banks

[5/] Only listed companies on SET (median); with revisions to data and methodoloty.

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